Day 3: Seasonal Forecast Application

Ghana Training Workshop

Katie Facer-Childs, Lucy Barker, Adriana Calderon, Eugene Magee, Ezra Kitson

29.01.26

NATIONAL CAPABILITY FOR GLOBAL CHALLENGES

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WASCAL
West African Science Service Centre on Climate Change and Adapted Land Use

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[Seasonal Forecasting Application]

Your team for today ☺

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Training Support

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Senior Hydroclimate Scientist
Workshop Lead

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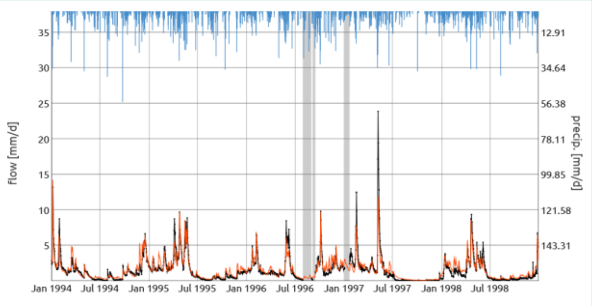
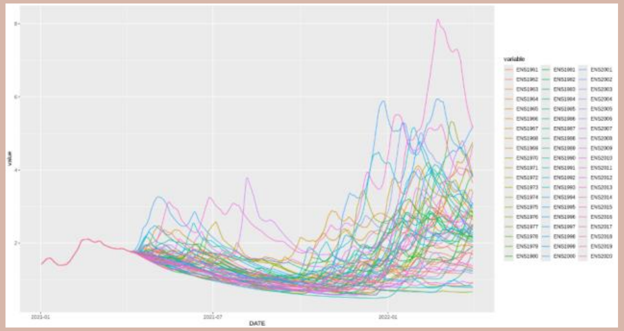
[Streamflow Forecasting Application]

Today's Exercises

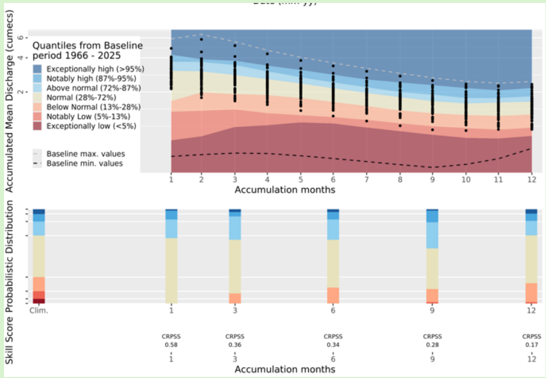
1
Calibrating and Running the GR6J Hydrological Model



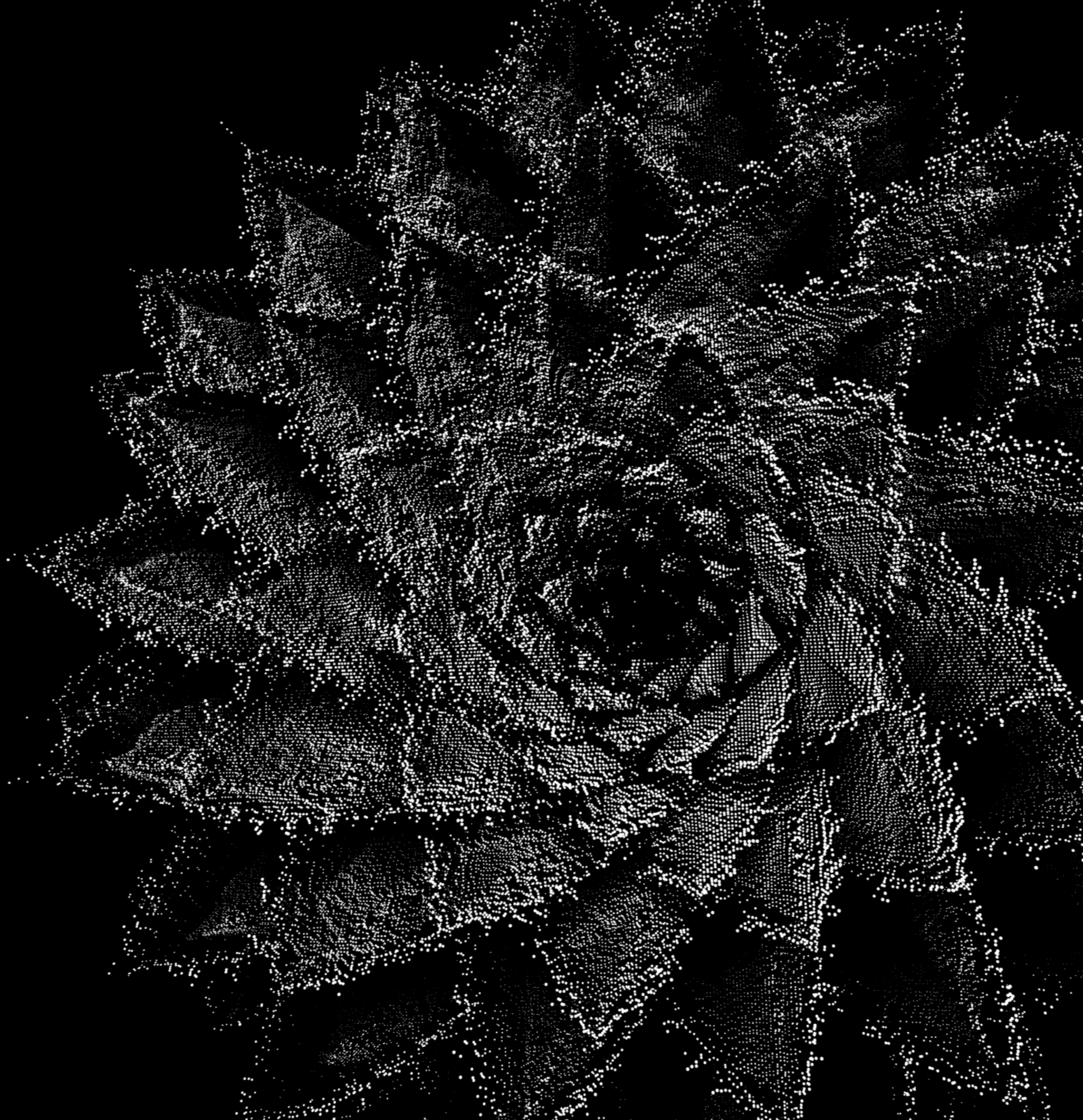
2
Running Ensemble Streamflow Prediction Forecasts



3
Visualising Forecasts for Decision Making



Hydrological Modelling



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[Hydrological Modelling]

Types of Hydrology Model

Increasing complexity and data requirements

Lumped Catchment Models

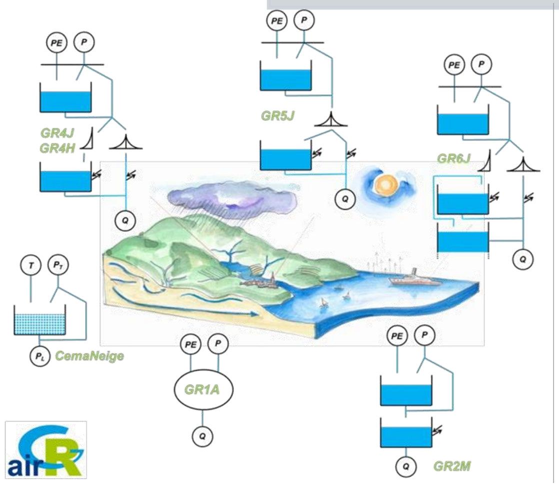
e.g.
- GR6J
- HBV
- SAC-SMA

Distributed Process-Based Models

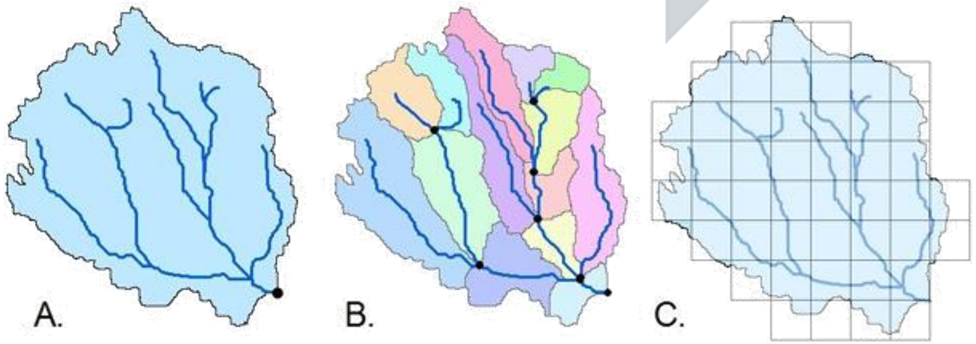
e.g.
- G2G
- SWAT
- MIKE SHE
- VIC

Coupled Earth-Atmosphere Land-Surface Models

e.g.
- JULES
- CLM
- NOAH-MP

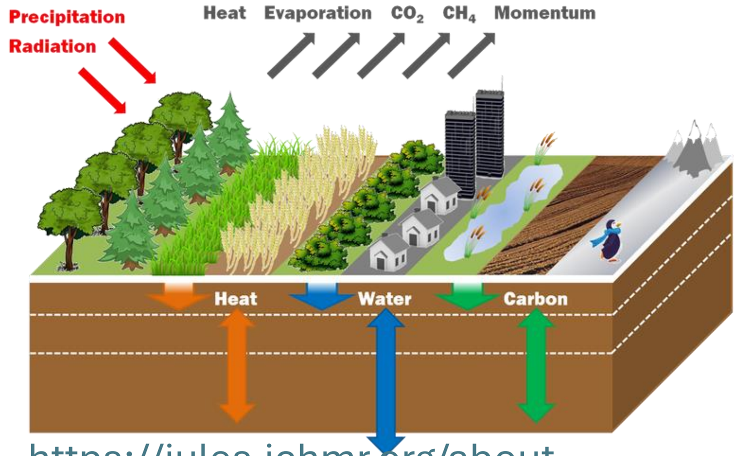


airGR: a suite of lumped hydrological models in an R-package



A. Lumped model
B. B. Semi-distributed model by sub catchment
C. Fully distributed model by grid cell

Sitterson et al. (2018) An Overview of Rainfall-Runoff Model Types



https://jules.jchmr.org/about

[Hydrological Modelling]

TODAY'S Model - GR6J

(Pushpalatha et al., 2011)

Simple lumped catchment model.

→ INPUT [catchment averaged] Rainfall and Potential Evapotranspiration (mm/day)

← OUTPUT river discharge (mm/day)

6 model parameters to calibrate:

X1 (max soil moisture storage),

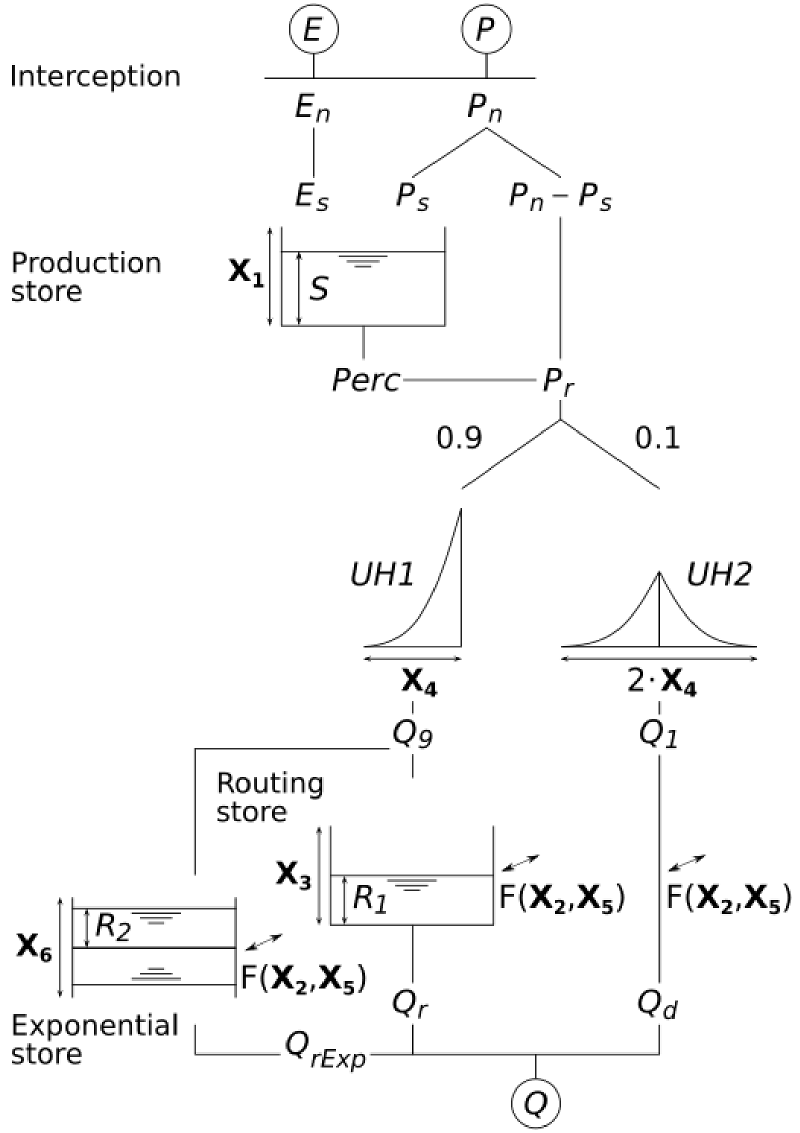
X2 (groundwater exchange coefficient),

X3 (max routing store capacity),

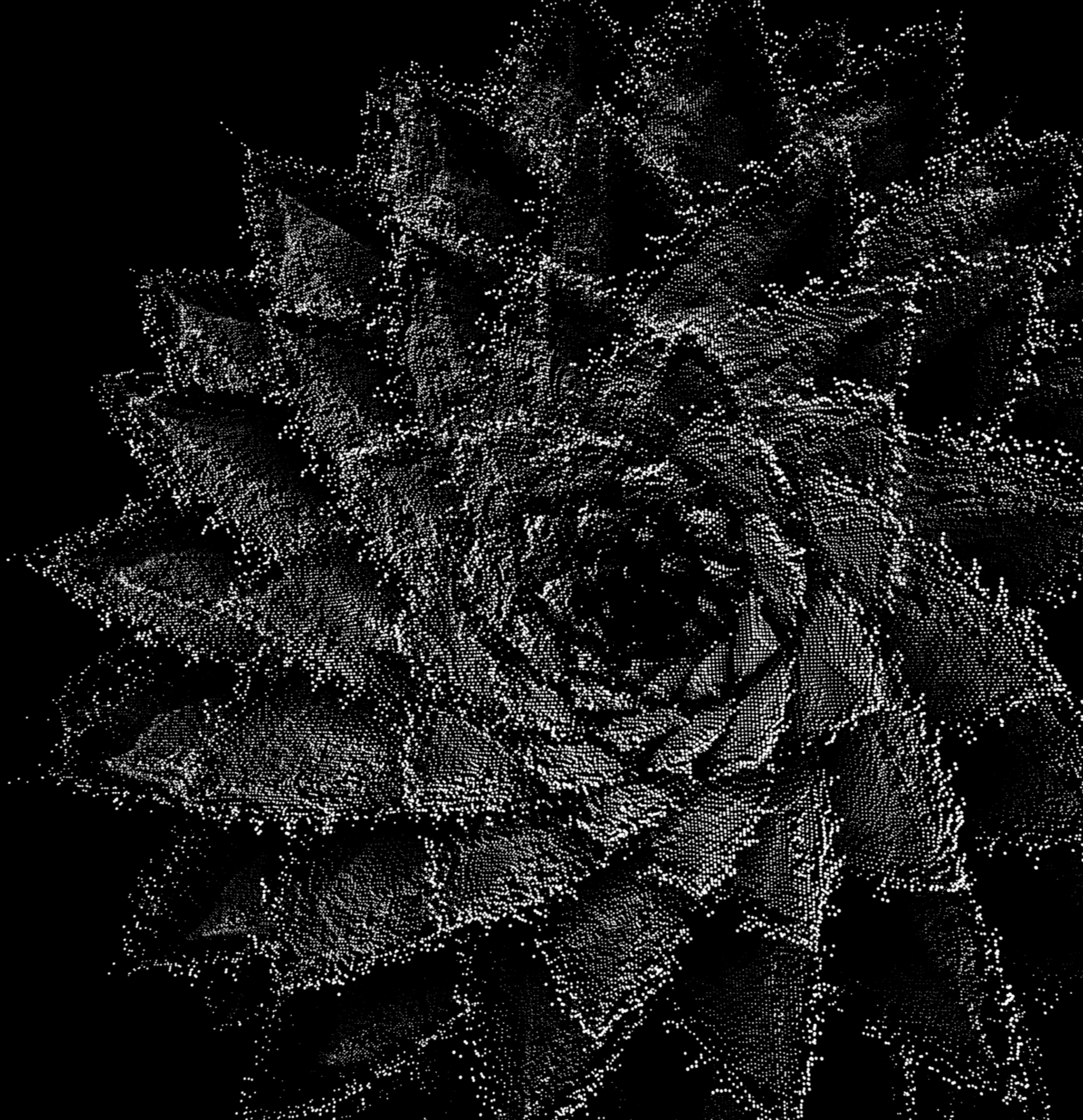
X4 (flow delay/peak time for unit hydrograph),

X5 (groundwater exchange threshold),

X6 (exponential storage coefficient)



Seasonal Forecasting



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[Seasonal Forecasting]

Why do a seasonal forecast?

An overview of the current hydrological status
including groundwater, river flow and soil moisture

An appraisal of where the current status is significantly different from 'normal'
For example indicating drought and flood susceptibility

An assessment of whether this is likely to get better or worse
over coming weeks and months

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[Seasonal Forecasting]

Why do a seasonal forecast?

Identifying both drought and flood rich periods for increased preparedness

Risk of hitting management thresholds

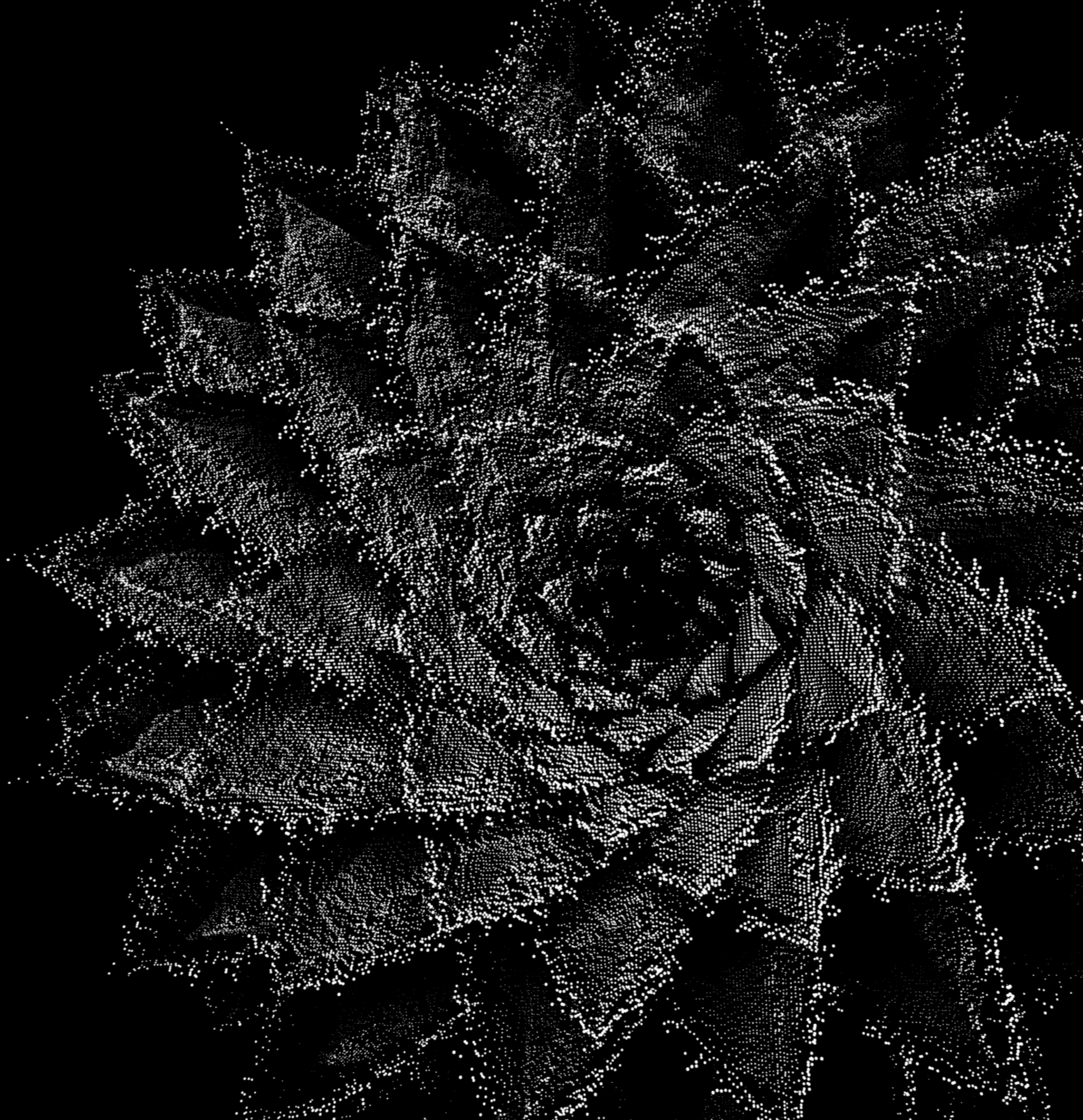
Aiding agricultural decision making e.g. irrigation needs

Manufacturing with high water needs

Drought impacts

Hydropower capacities

Seasonal Forecasting Methods



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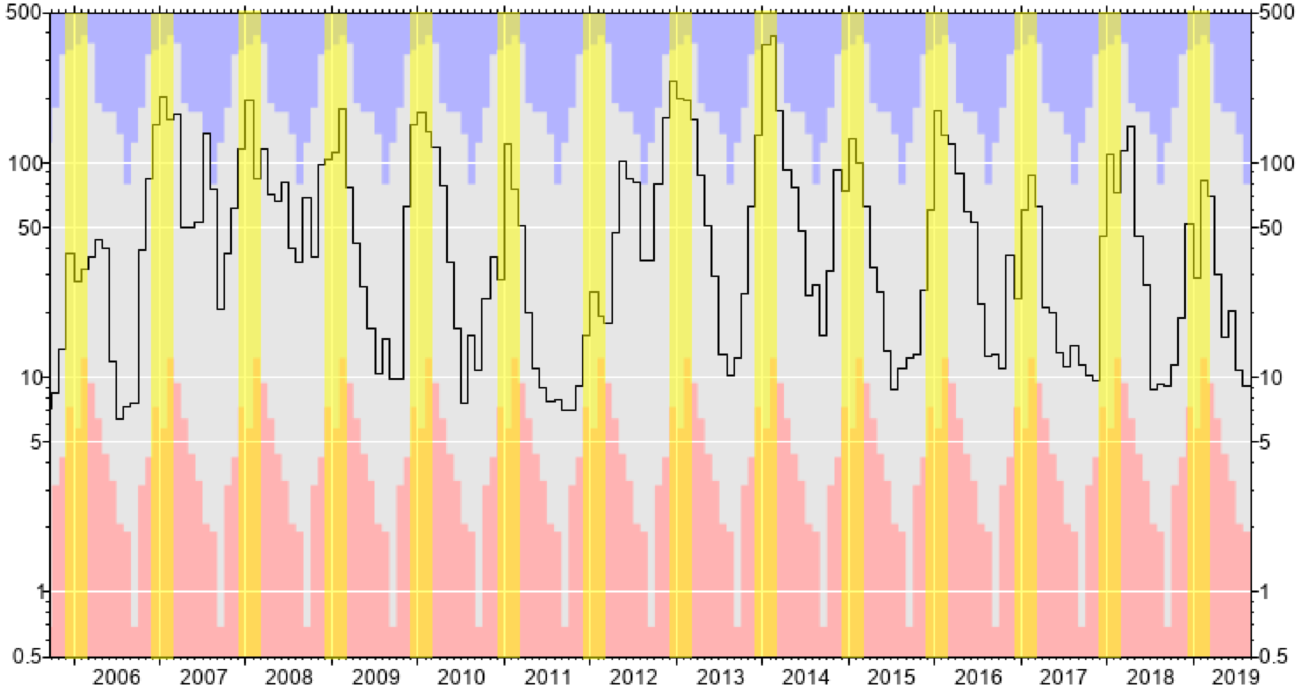
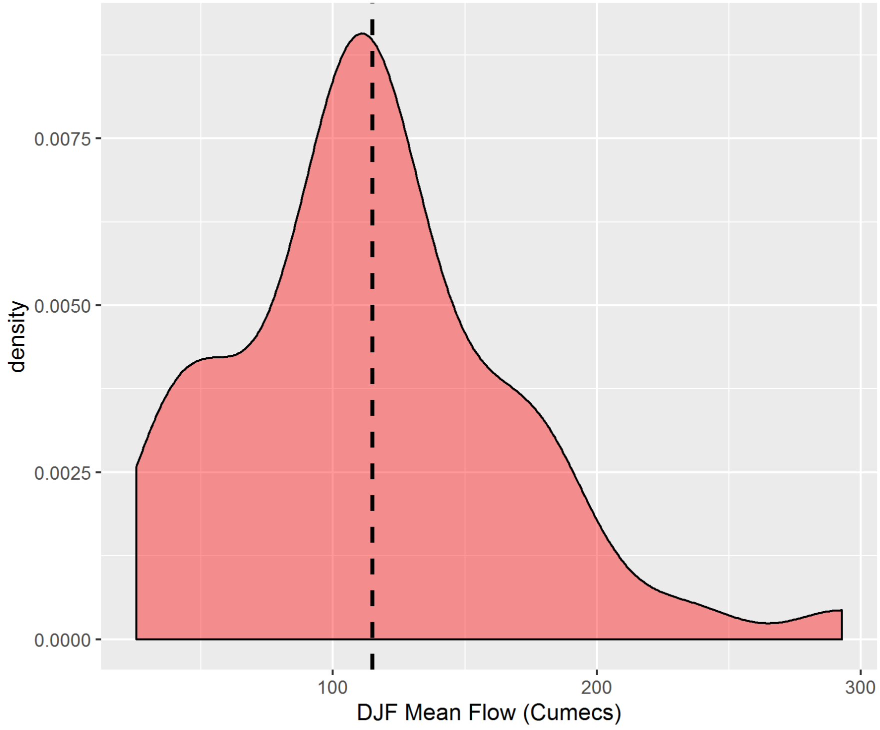
[Seasonal Forecasting Methods]

Climatology

Simplest method of forecasting.

Uses only streamflow data.





Provides a probability distribution of flows.

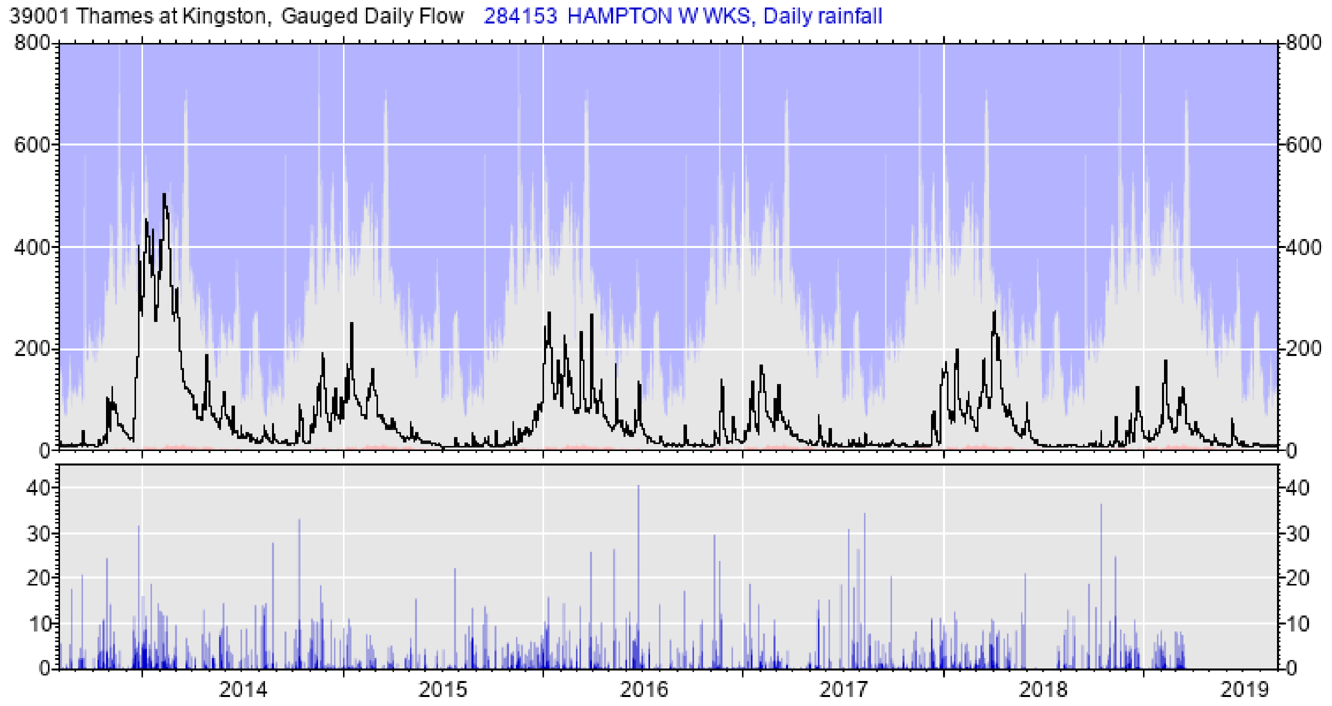
Can calculate deterministic mean if required.

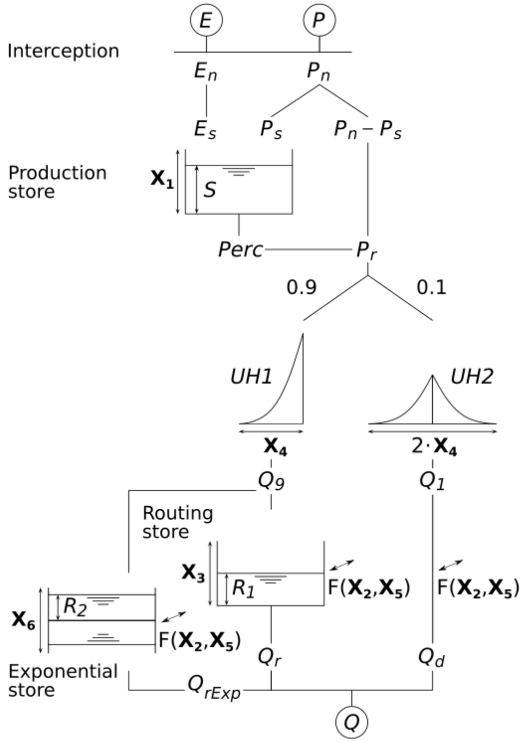
[Seasonal Forecasting Methods]

Ensemble Streamflow Prediction

A step further than Climatology

We know the initial conditions of the catchment





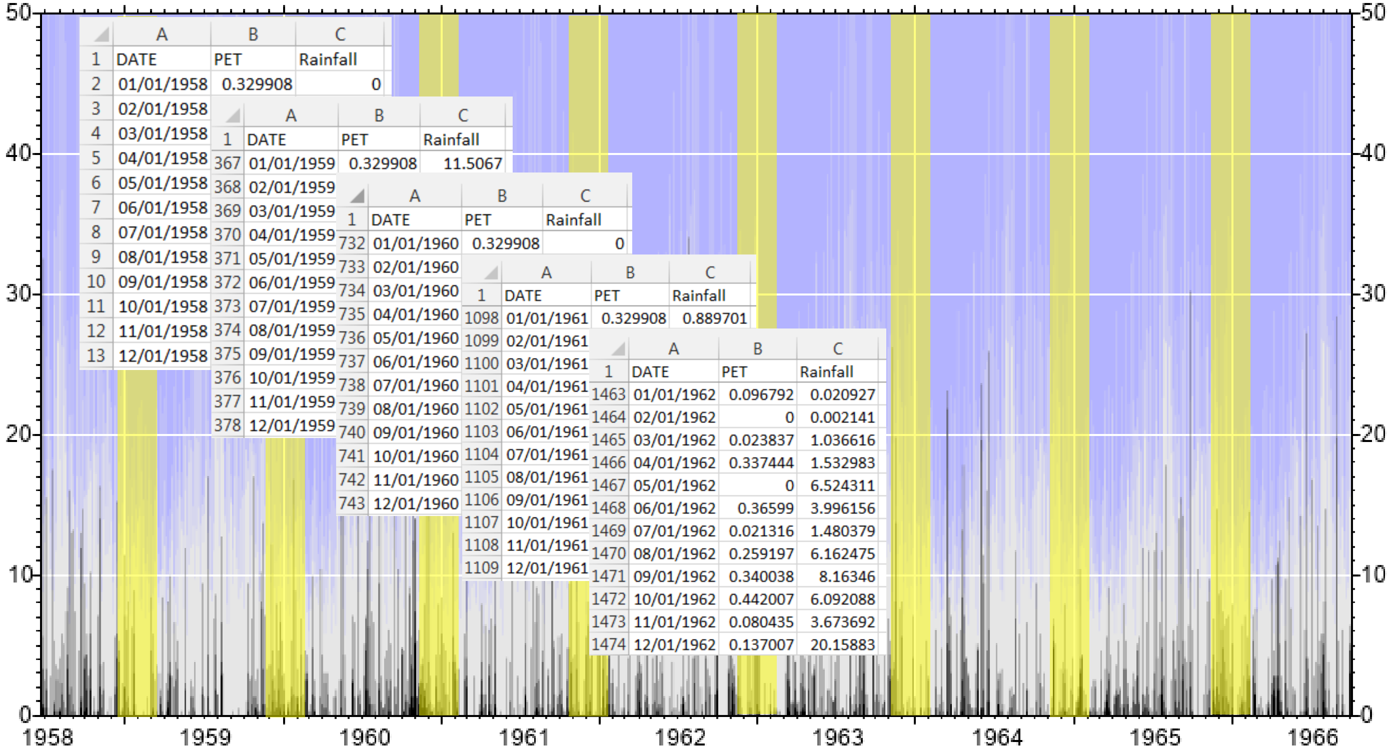
Model flow until the present day using observed climate data

(Rainfall and PET)

[Seasonal Forecasting Methods]

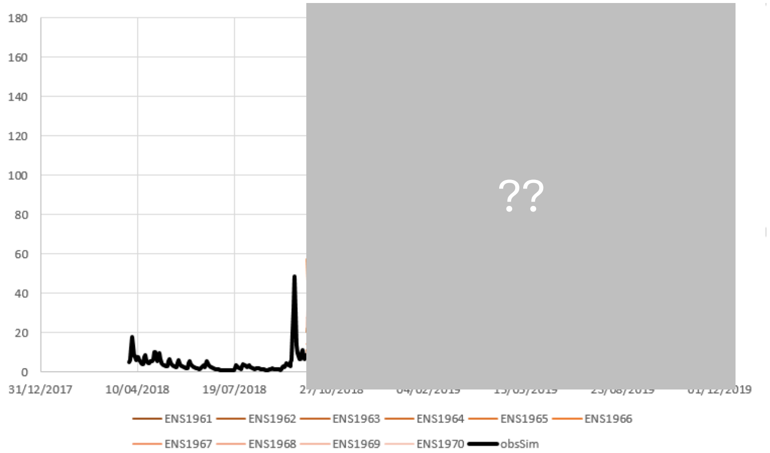
Ensemble Streamflow Prediction

Then drive the model with climate data from each historic year

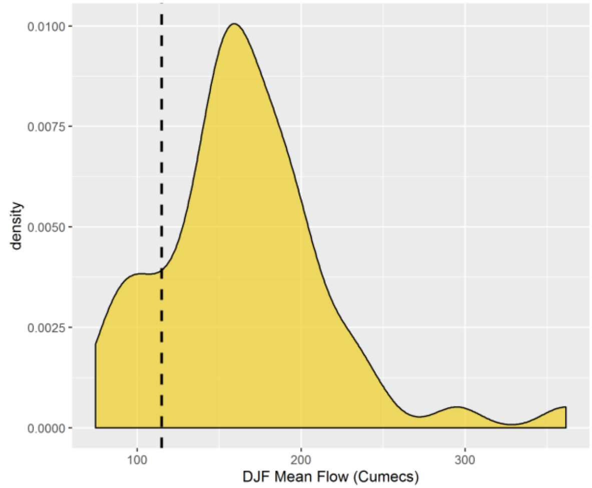


Initial Streamflow Conditions

Ensemble of Streamflow Forecasts



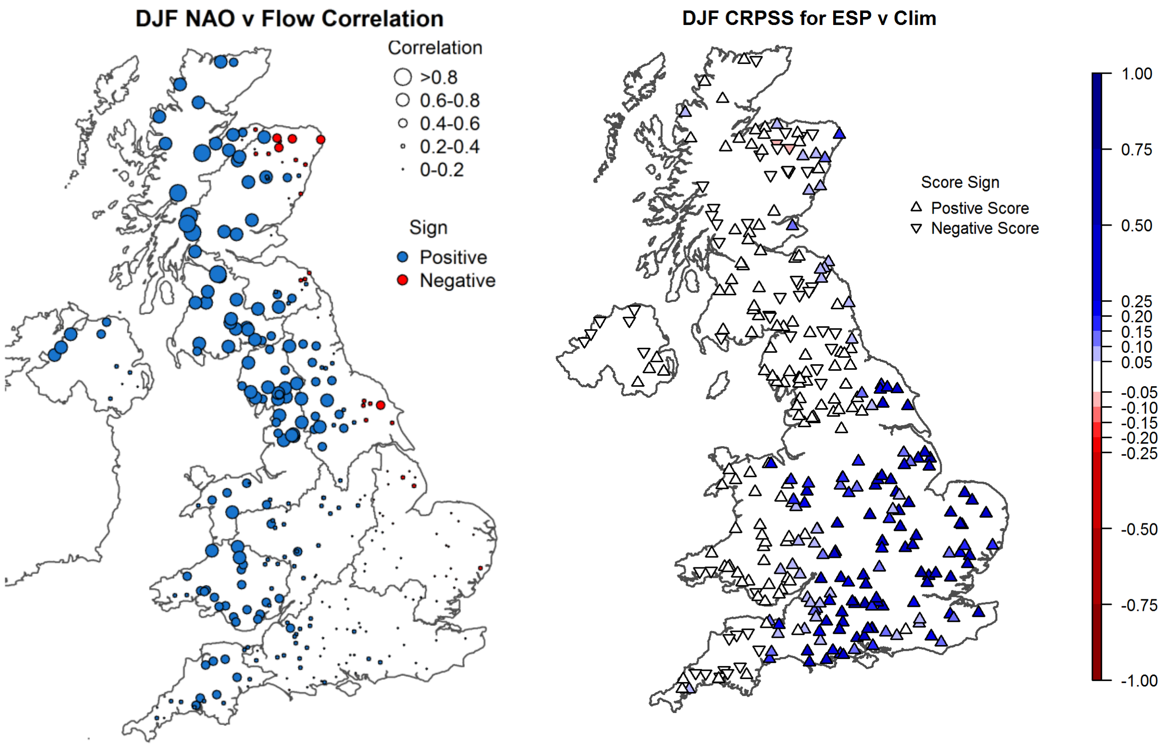
Provides a probability distribution of mean flow

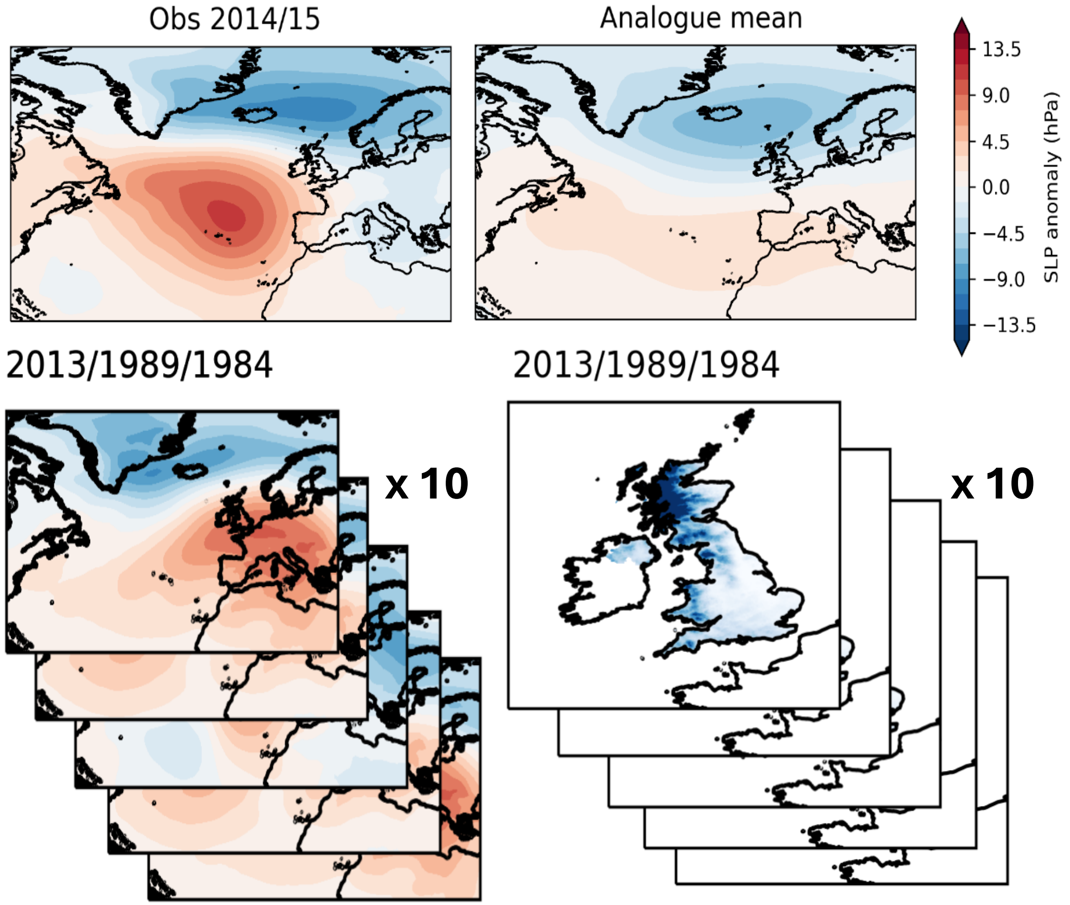


Dashed line = mean of climatology

[Seasonal Forecasting Methods]

Weather Analogues (selective ESP)





10 variants x 51 members

[Seasonal Forecasting Methods]

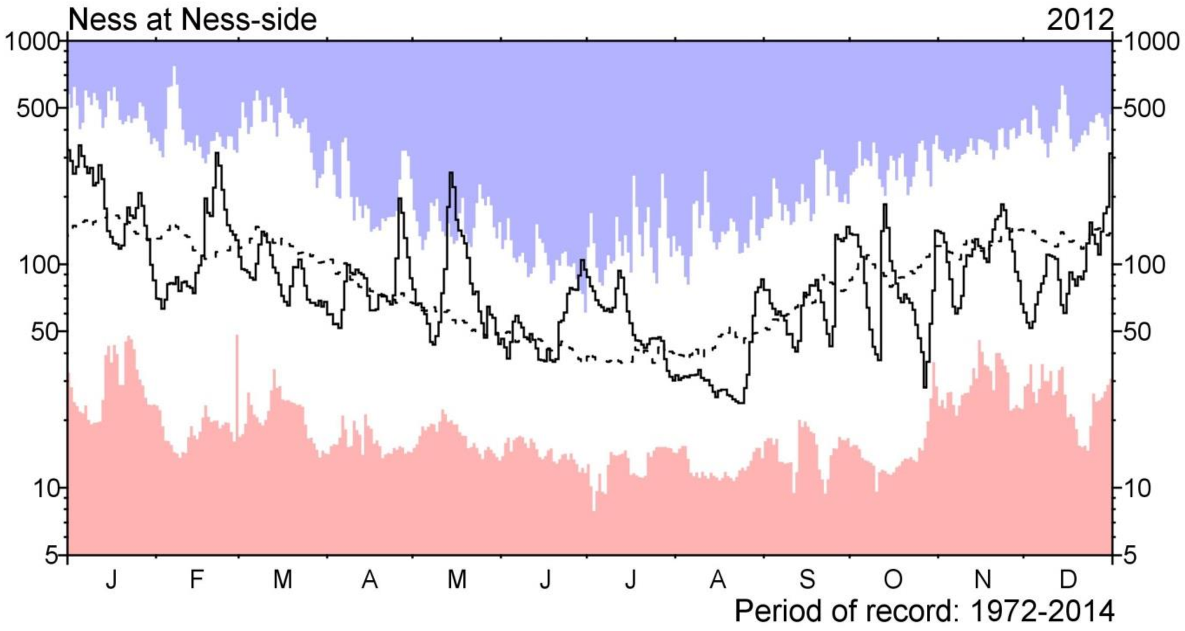
Persistence and Analogues

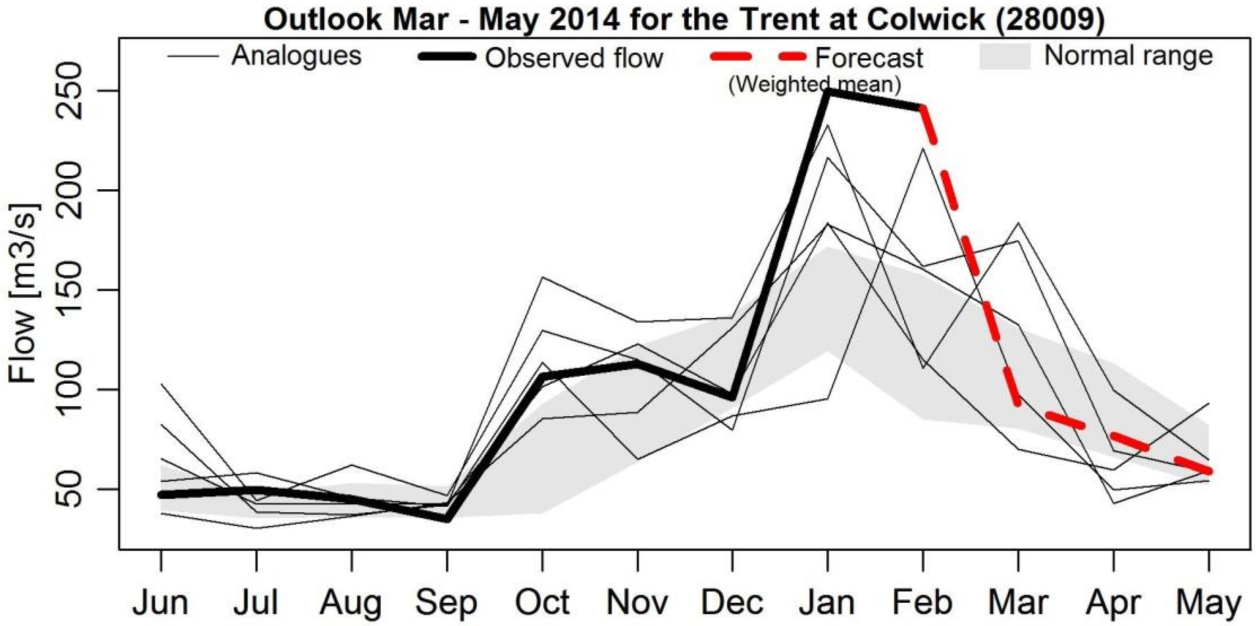
Persists the flow anomaly from one month to the next.

Using standardised anomalies reflects the seasonal cycle

Five analogues chosen from the past flow record

Assumes similarity in the past will indicate the future





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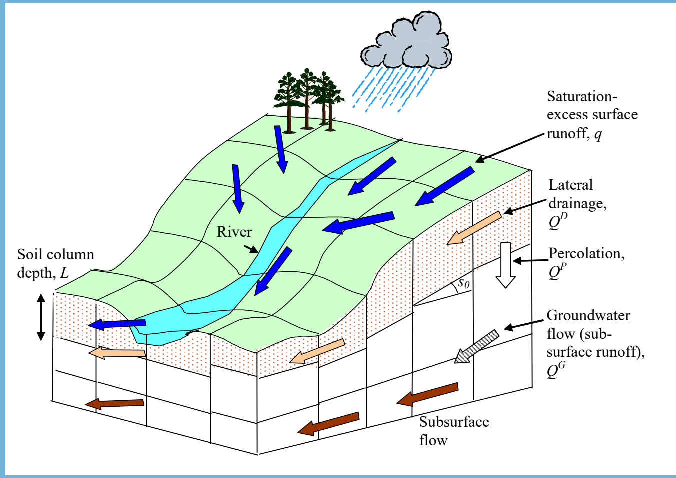
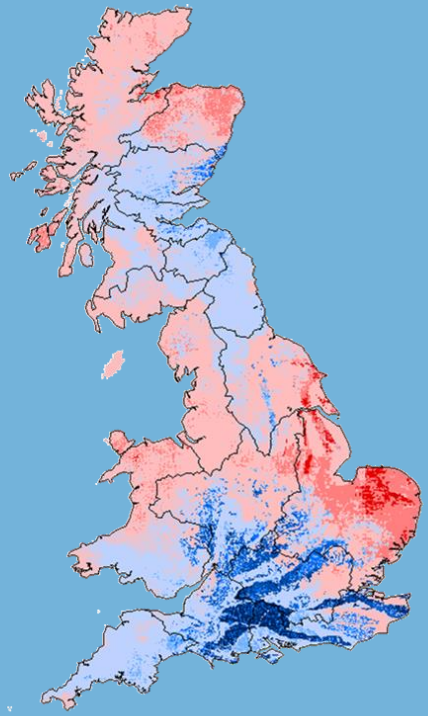
[Seasonal Forecasting Methods]

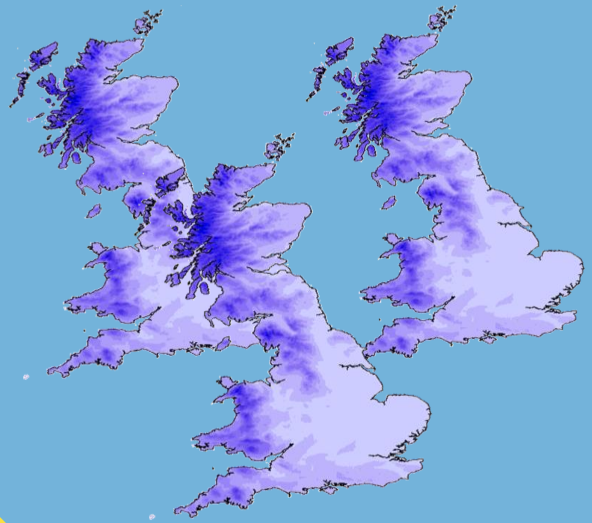
Dynamic Forecasts



G2G run with observed rainfall and PET

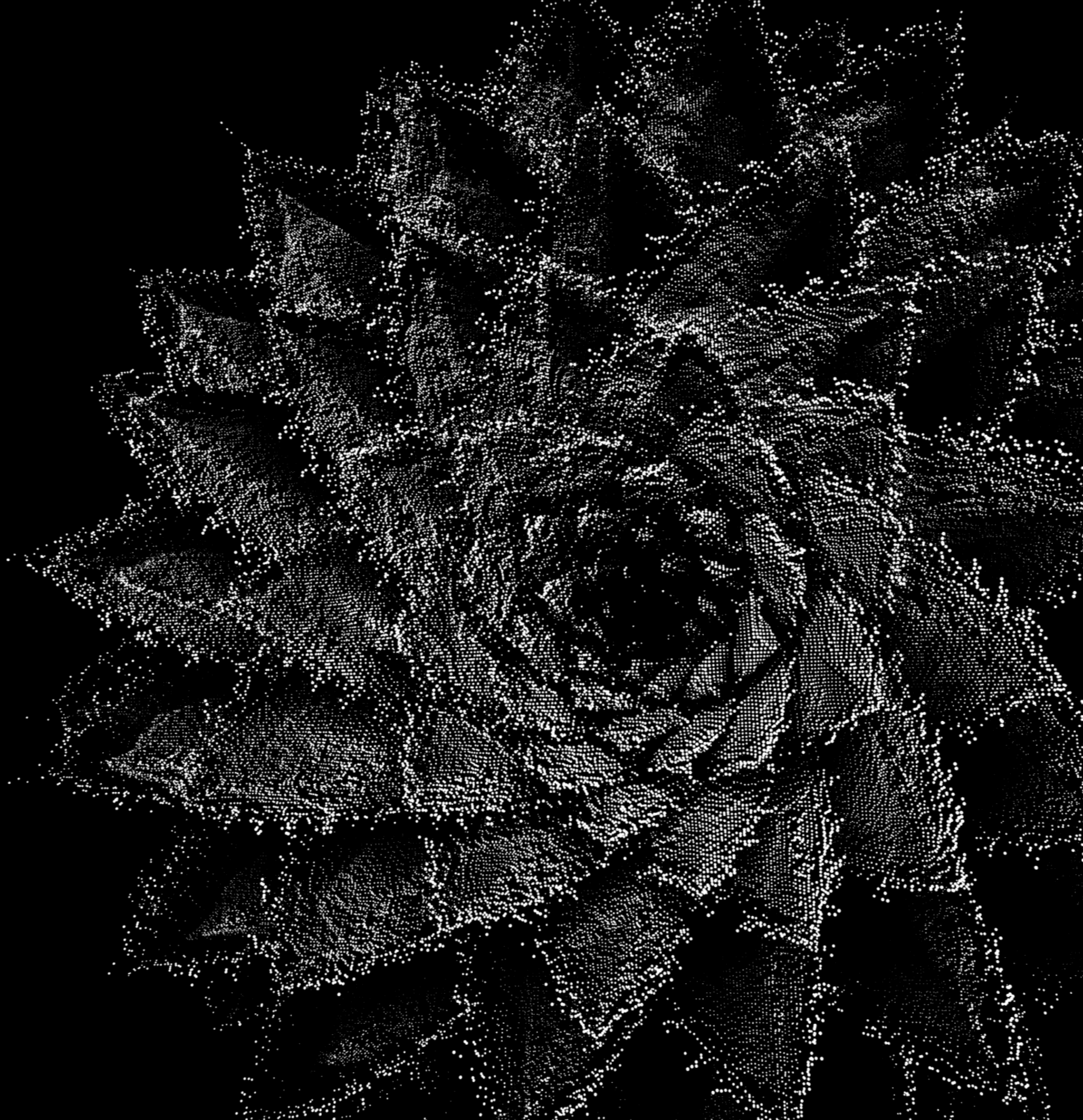


Initial Hydrological Conditions



Rainfall Forecasts + mean PET run through model

Calculating Skill



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[Calculating Skill]

Running a Reforecast/Hindcast

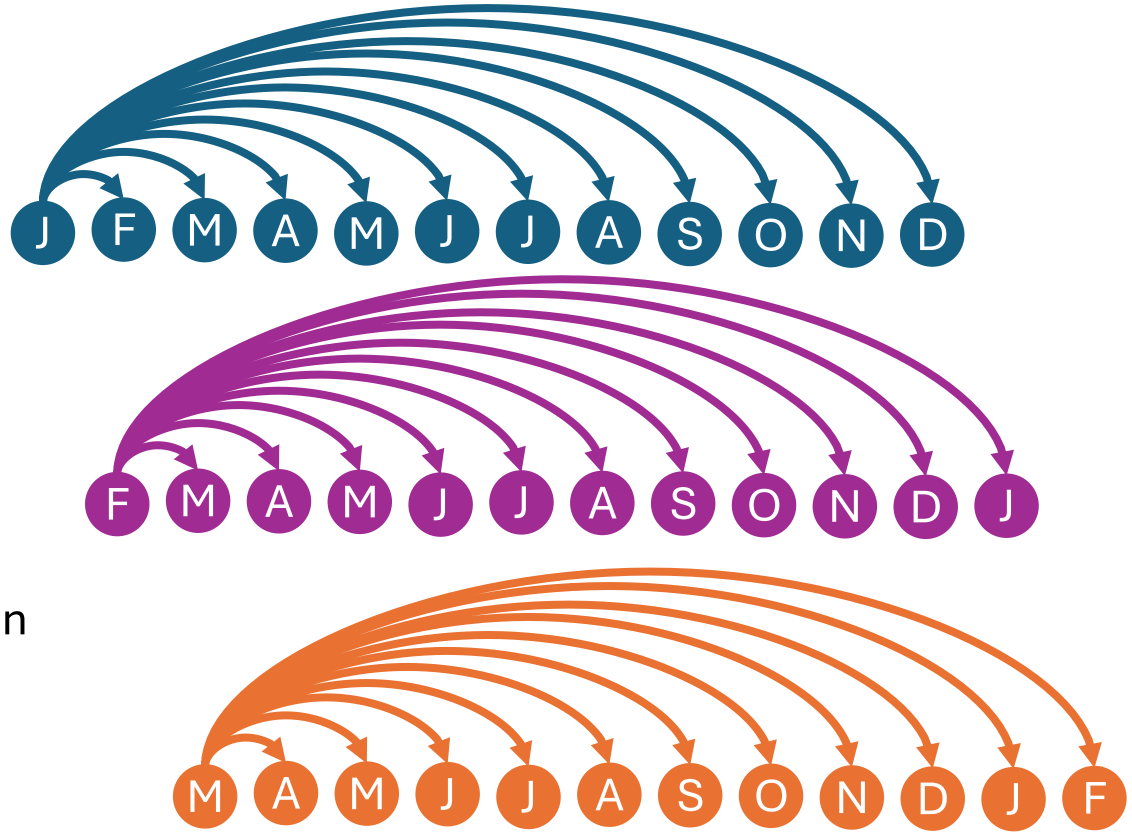
Forecast skill is different to model performance.

Model performance = can my model simulate flow?
Model skill = can my model predict what will happen next?

In order to calculate the skill of your forecasts you need to run a “reforecast”, or “hindcast”.

HINDCAST / REFORECAST = FORECAST of the PAST

Even for a simple model like GR6J, this can be quite computationally intensive – so we won’t do this today, we will use pre-calculated skill scores in our notebooks.



Run all lead times, for each initialisation month, for each starting month over your hindcast period (e.g. 20 years = 240 ESP forecasts = 2880 months of forecast data)

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[Calculating Skill]

You know the flow over your hindcast period, so you can calculate how well your model did.

Skill Scoring

Continuous Ranked Probability Skill Score (CRPSS)

Commonly uses climatology as a “benchmark” – how much better is the forecast than if we use climatology?

R/Python packages that will calculate it. e.g. easyVerification

The Equation

The formula for the CRPSS is:

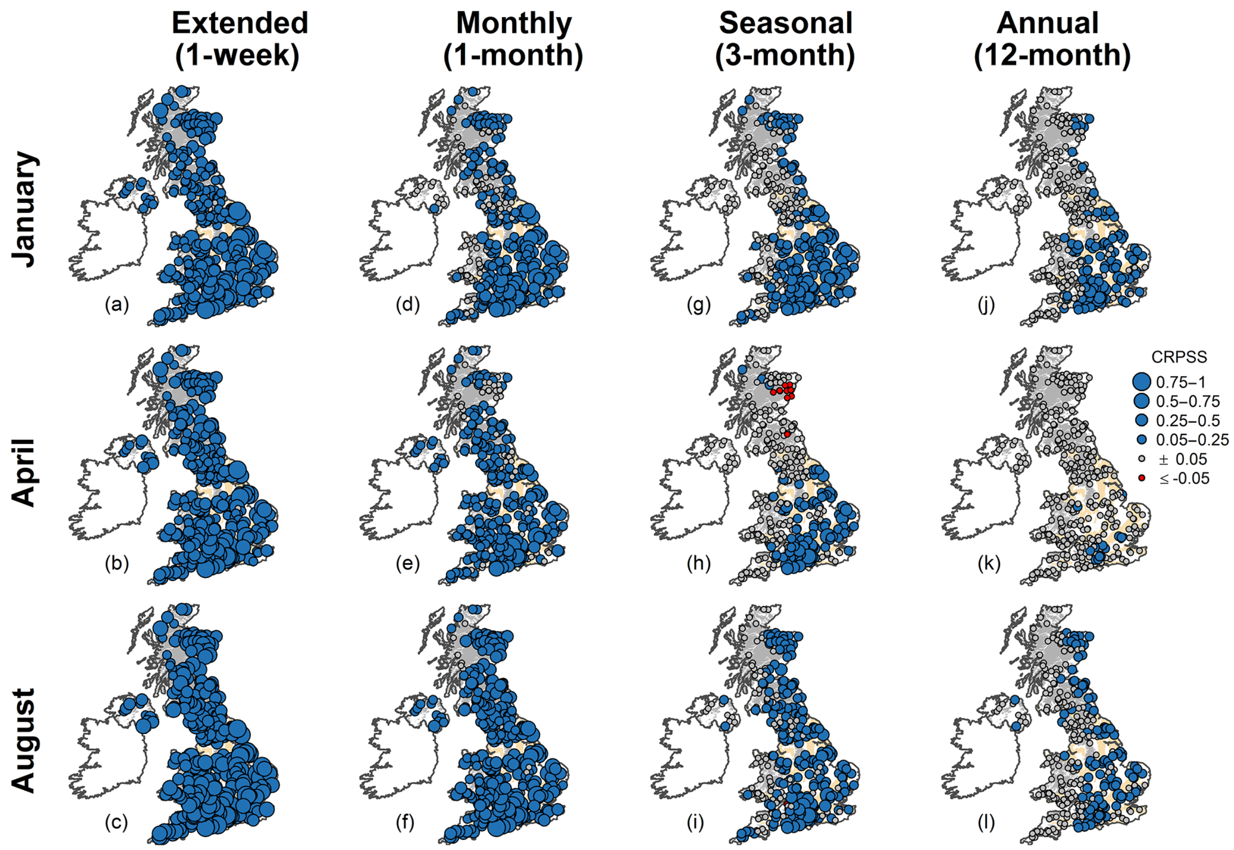
$$\text{CRPSS} = 1 - \frac{\overline{\text{CRPS}}_f}{\overline{\text{CRPS}}_{ref}}$$

Where:

- $\overline{\text{CRPS}}_f$ is the average Continuous Ranked Probability Score (CRPS) of the forecasting system.
- $\overline{\text{CRPS}}_{ref}$ is the average CRPS of the reference forecast (e.g., climatology).

Interpretation

- $\text{CRPSS} = 1$: Perfect forecast (the system has 100% skill over the reference).
- $\text{CRPSS} > 0$: The forecasting system is more accurate than the reference.
- $\text{CRPSS} = 0$: The system has no skill compared to the reference.
- $\text{CRPSS} < 0$: The system is less accurate than the reference.



Harrigan et al., (2018) Benchmarking ESP skill in the UK

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[Calculating Skill]

Do I need to do this?

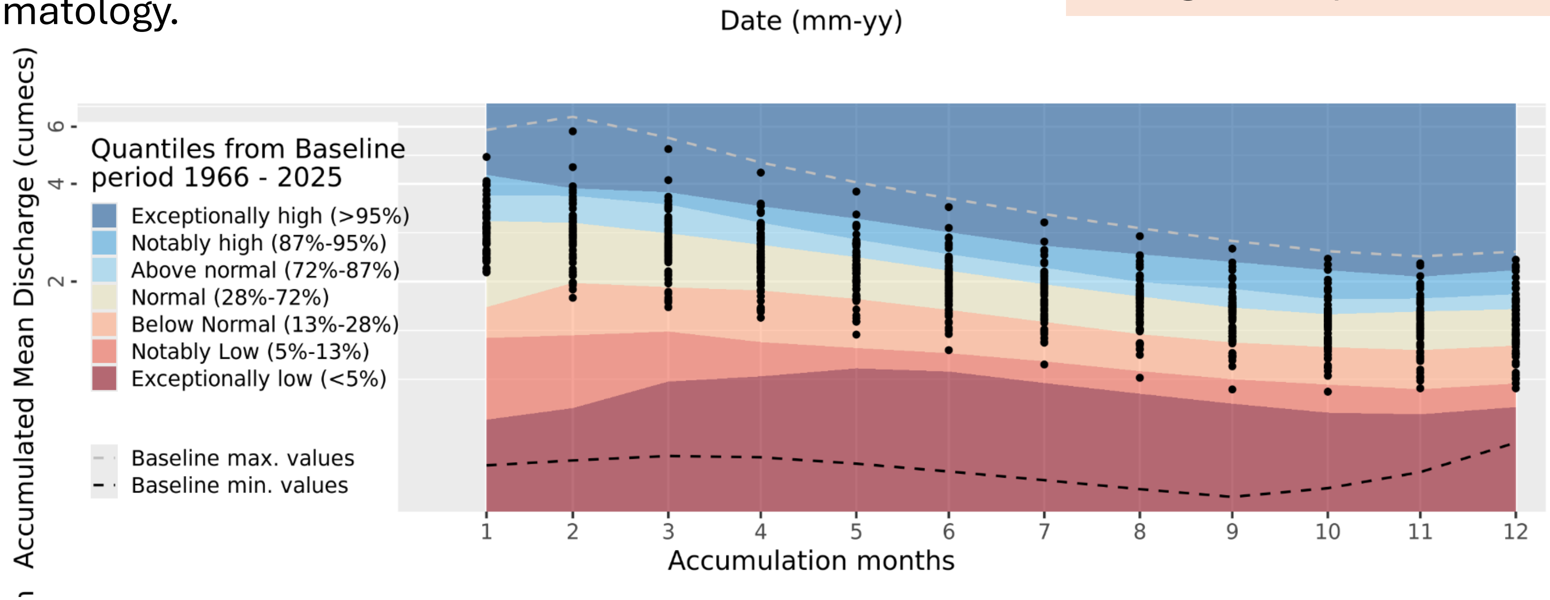
ESP is a simple forecasting method using observed climate.
If there is no (or no longer) an impact of the initial conditions on your river flows, then ESP defaults to climatology.

Assuming your model PERFORMANCE is good (the model is good at simulating observed flows), then your ESP forecast will be no worse than climatology.

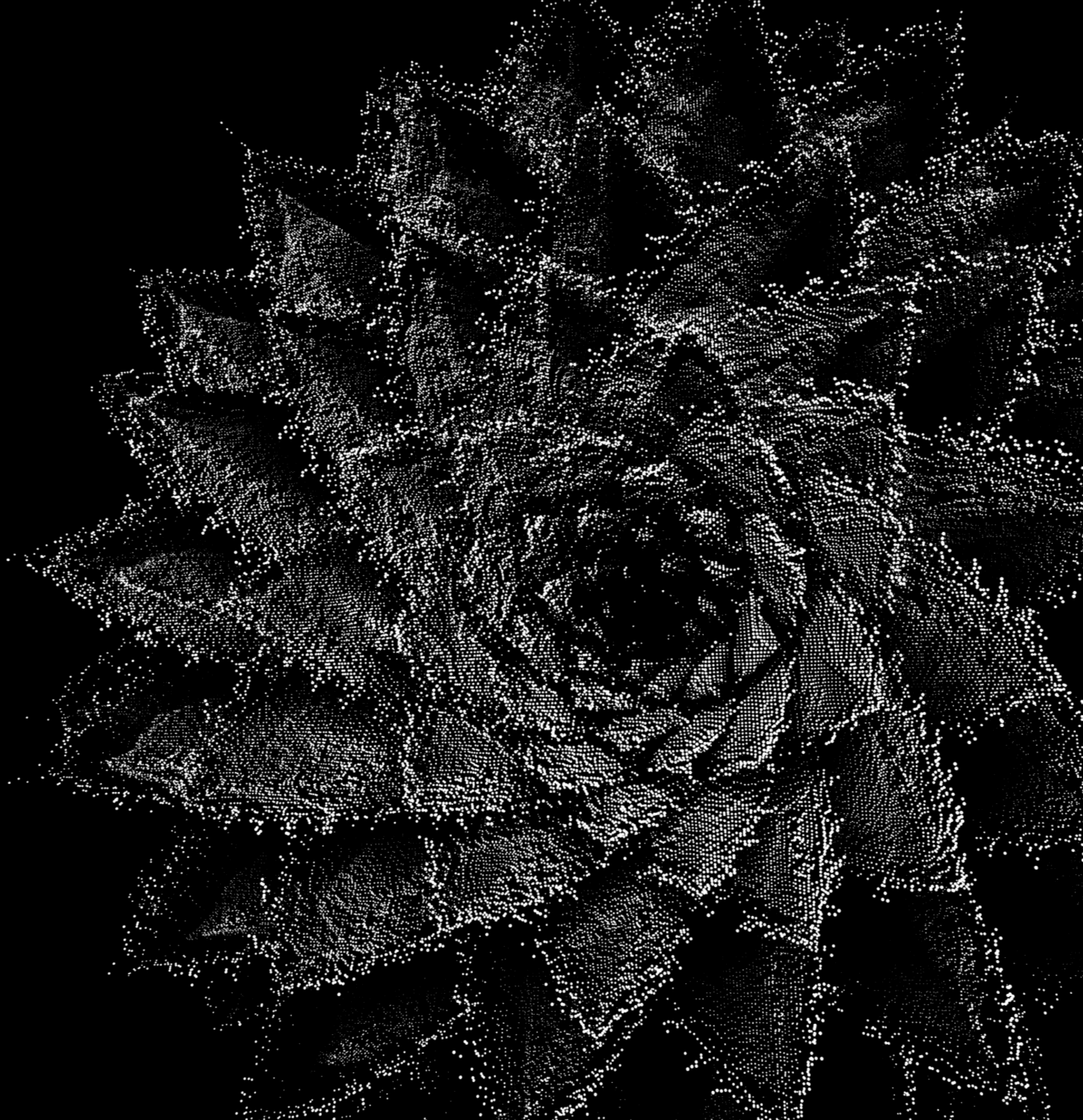
You can expect that your ESP forecast will be better than climatology in catchments with high storage (if its not artificially influenced)

If your forecast uses a meteorological forecast models' outputs, your skill could be worse than climatology if the met model is not good!

Hindcasts should be rerun if you change your forecast system (e.g. recalibrate the model or change the input data source)



Visualising Outputs



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[Visualising Outputs]

Common Categorisation

7 categories

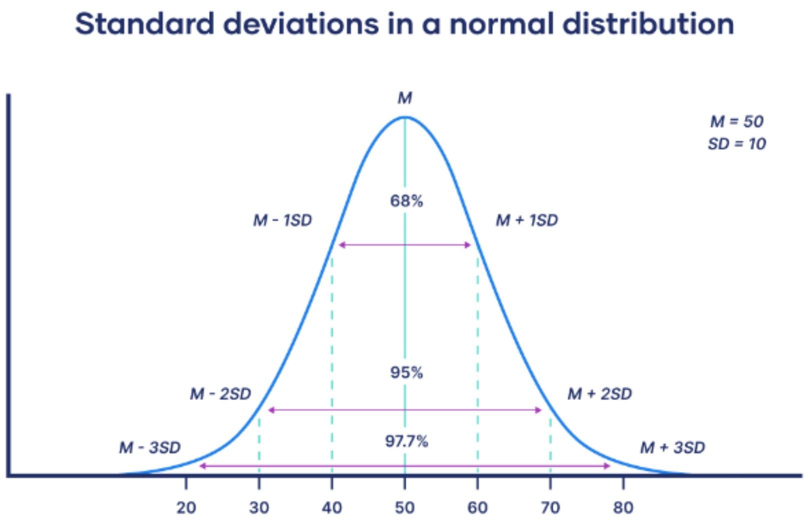
- Exceptionally high >95%
- Notably high 87-95%
- Above normal 72-87%
- Normal 28-72%
- Below normal 13-28%
- Notably low 5-13%
- Exceptionally low <5%

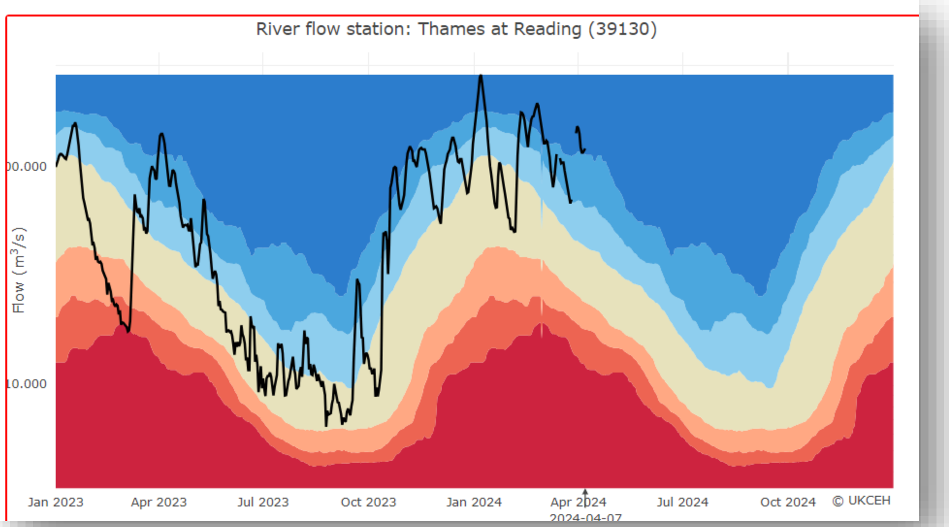
5 categories

- High >87%
- Above normal 72-87%
- Normal 28-72%
- Below Normal 13-28%
- Low <13%

3 categories

- Above normal >72%
- Normal 28-72%
- Below normal <28%





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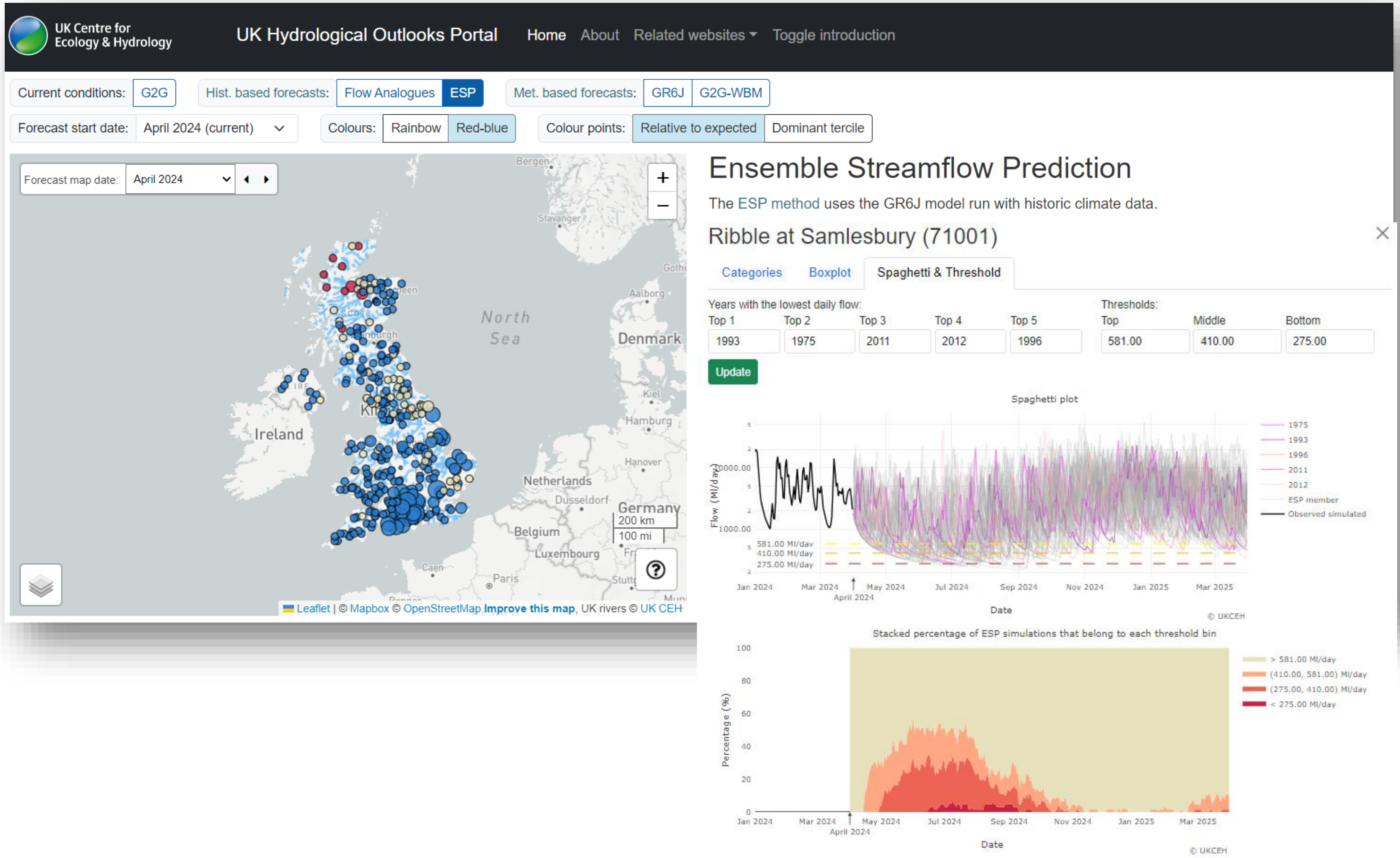
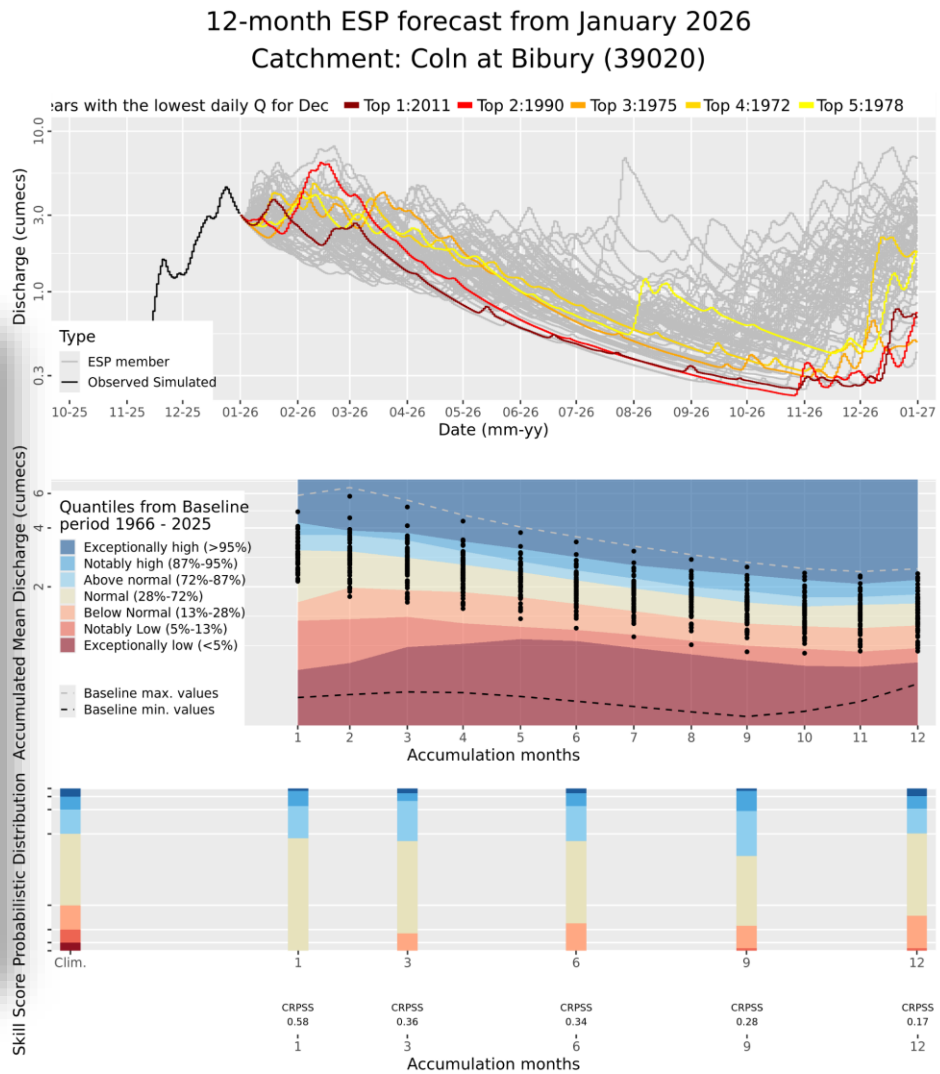
[Visualising Outputs]

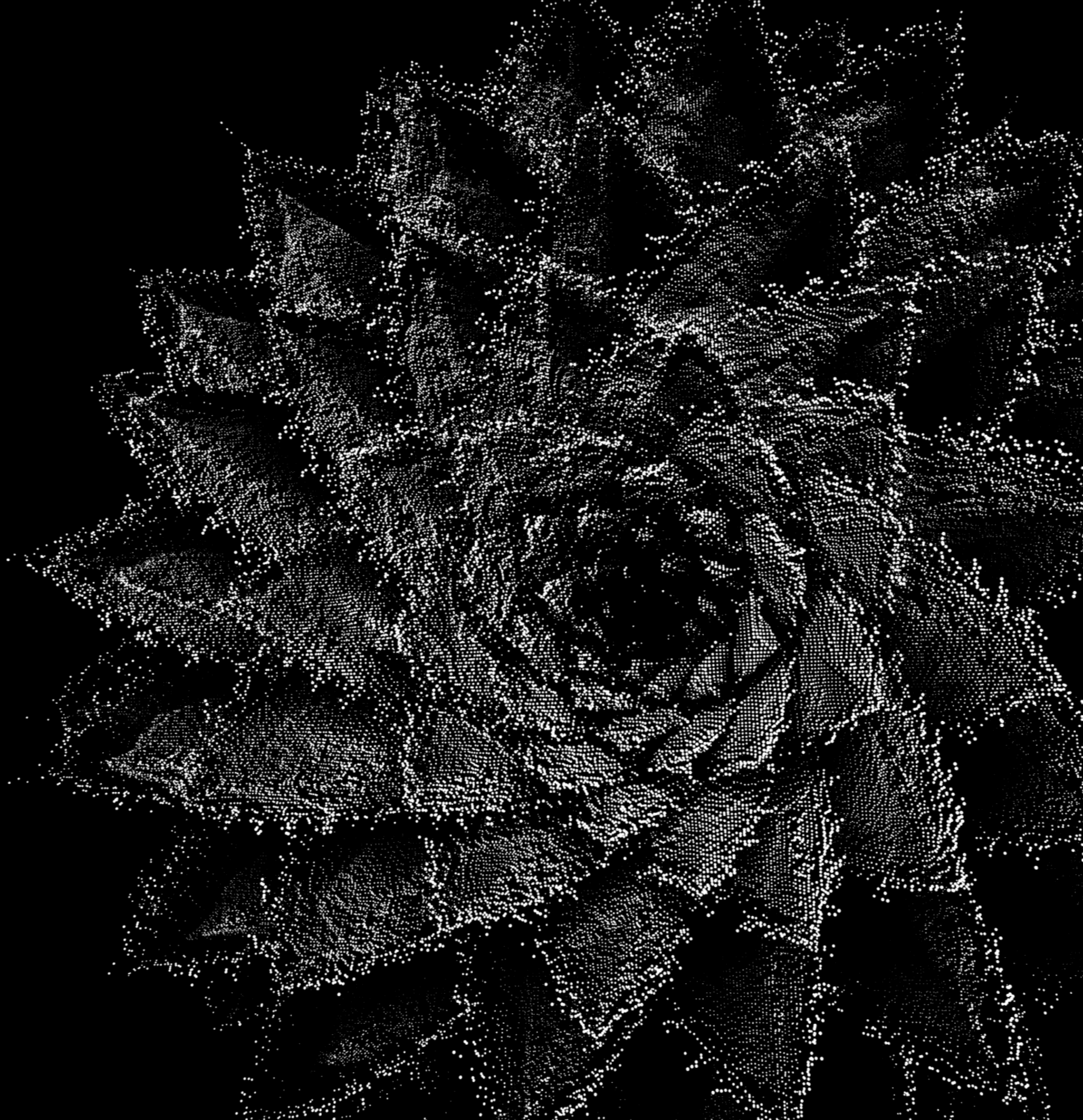
Graphics for Decision Making





The Continuous Ranked Probabilistic Skill Score (CRPSS) is one of many scores used by forecasters to assess ho
(how good) a forecasting system is. It tells us how much better the forecast is at predicting streamflow compar
benchmark (here: the climatology or long term average). The degree of skill based on CRPSS: very high [0.75,
high [0.5, 0.75); moderate [0.25, 0.5); low (0, 0.25); no skill=0; and negative skill<0 (meaning climatology is a
of what is going to happen than the forecast). To find out how these skill scores were derived, see Harrigan et a
https://www.hydrol-earth-syst-sci.net/22/2023/2018/hess-22-2023-2018.html

Operational Seasonal Forecasting



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[Operational Seasonal Forecasting]

The UK Hydrological Outlook

Provides UK-wide seasonal (1- and 3-month) forecasts of river flow and groundwater levels

Monthly, operational since summer 2013

Three modelling methods for hydrology:

1. Persistence and Analogy,
2. Ensemble Streamflow Prediction,
3. Dynamic Rainfall Forecast

Freely available: http://www.hydoutuk.net/

Prudhomme, et al. (2018). Hydrological Outlook UK: an operational streamflow and groundwater level forecasting system at monthly to seasonal time scales
Hydrological Sciences Journal

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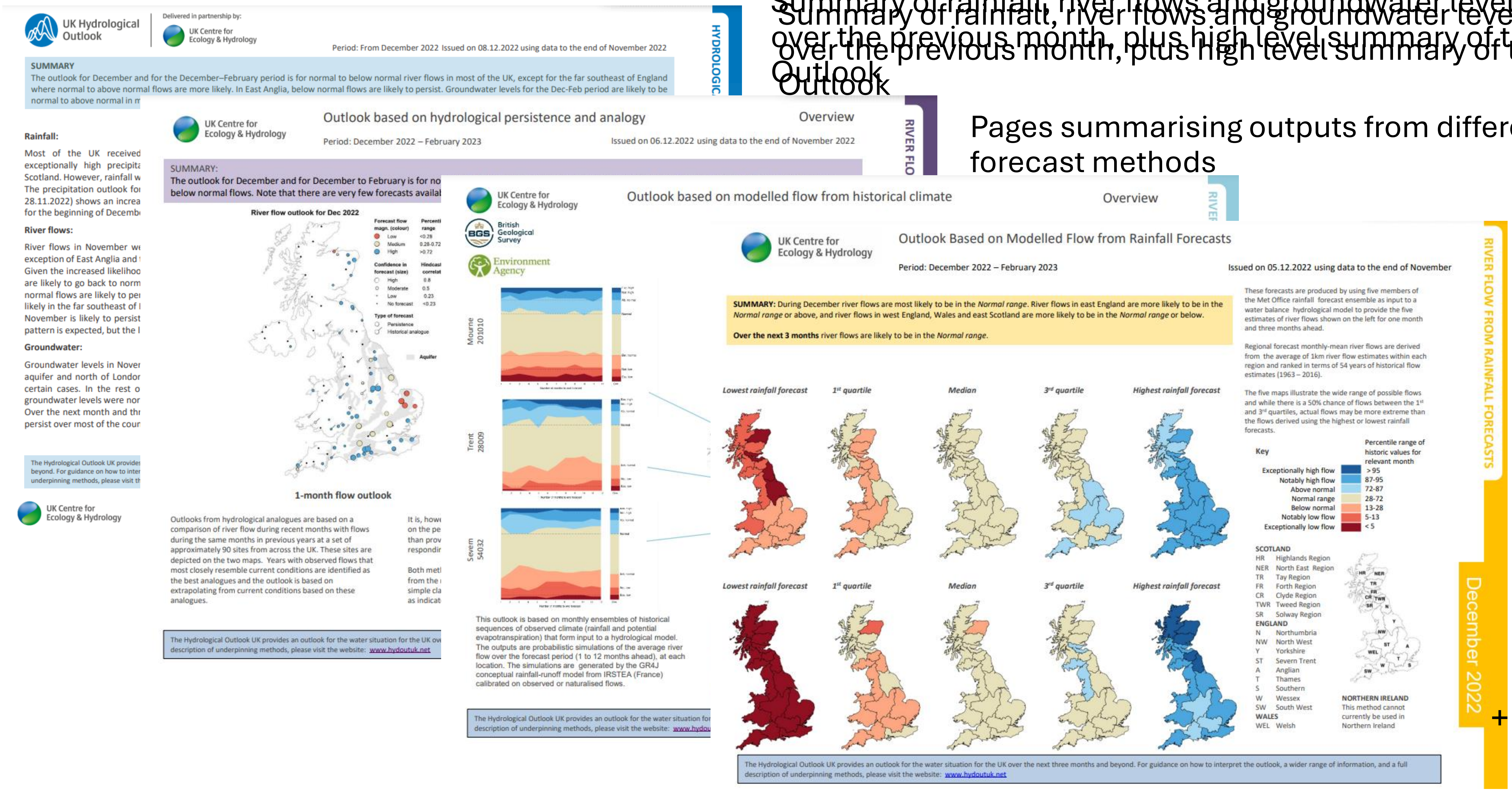
Summary of rainfall, river flows and groundwater levels over the previous month, plus high level summary of the Outlook

Summary of rainfall, river flows and groundwater levels over the previous month, plus high level summary of the Outlook



Pages summarising outputs from different forecast methods

+ more!

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[Operational Seasonal Forecasting] **The UK Hydrological Outlook**

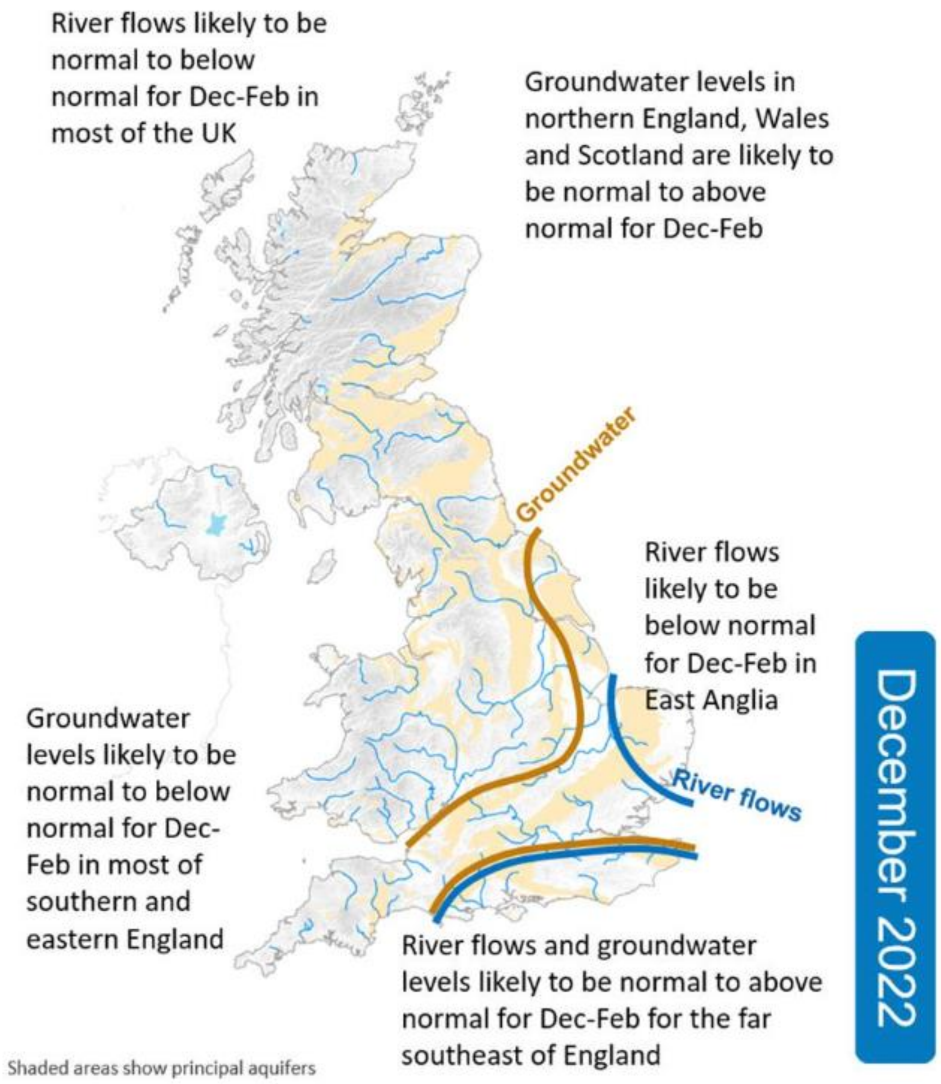
Method	UK best skill (seasons)	UK best skill (location)	UK Reference	Summary
1. Persistence and analogy	At 1 to 3mn lead time, most skilful for **Summer**	South and East	Svensson (2016)	**Simple, just needs past river flow data (no met. forecasts)**
2. Ensemble Streamflow Prediction (ESP)	At 3 to 12mn lead time, most skilful for **Autumn/Winter** At 1mn lead time, most skilful for **Summer**	South and East	Harrigan et al. (2018)	**Fairly simple – freely available, past climate and river flow data**
3. Flows modelled using dynamic rainfall ensemble	At 1 to 3mn lead time, most skilful for **Autumn/Winter**	North and West	Bell et al. (2017)	**Complex – needs weather forecasts, complex distributed model (but could use a simpler one)**

Increasing complexity of modelling and data requirements

[Operational Seasonal Forecasting]

The UK Hydrological Outlook – combing approaches

- Multiple approaches to capitalise on strengths of different methods in space and time
- It can be challenging to combine outputs over different spatial and temporal scales
- Monthly discussion forum with all modelling teams in the room to identify the signals emerging across all approaches



Summary map presents key messages for the forecast

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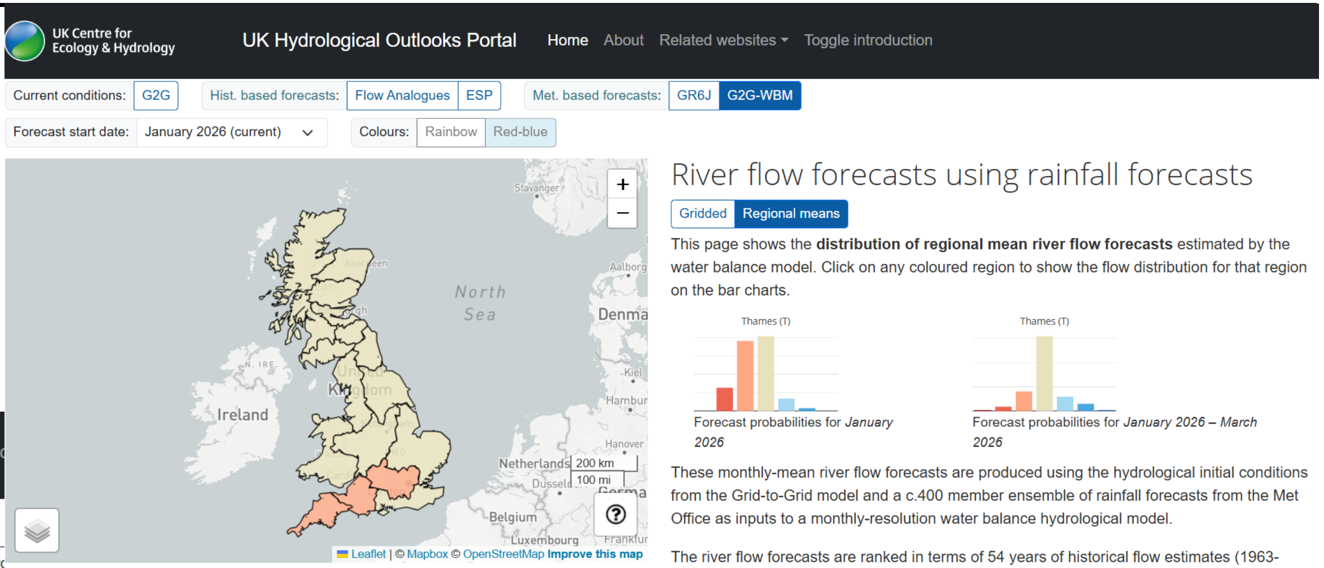
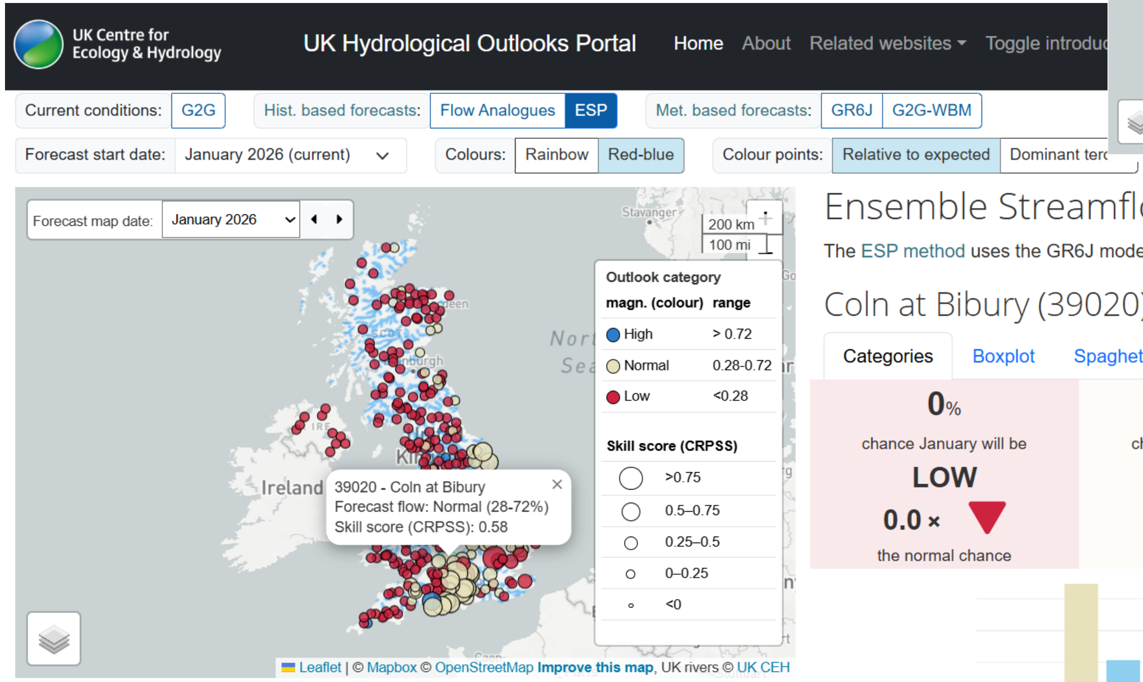
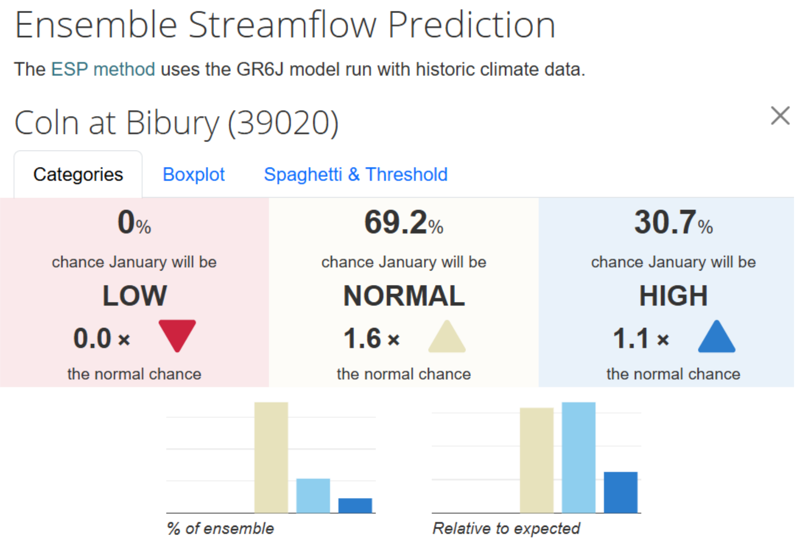
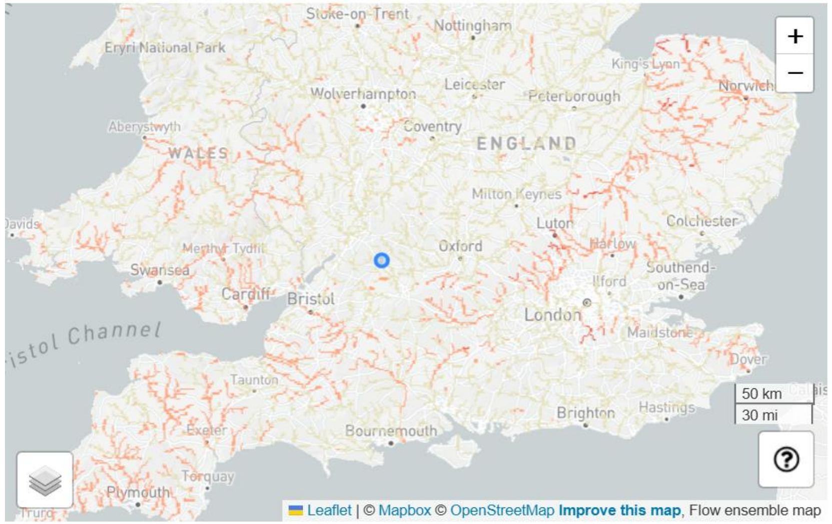
[Operational Seasonal Forecasting]

The UK Hydrological Outlook









Interactive model outputs available at www.ukho.ceh.ac.uk

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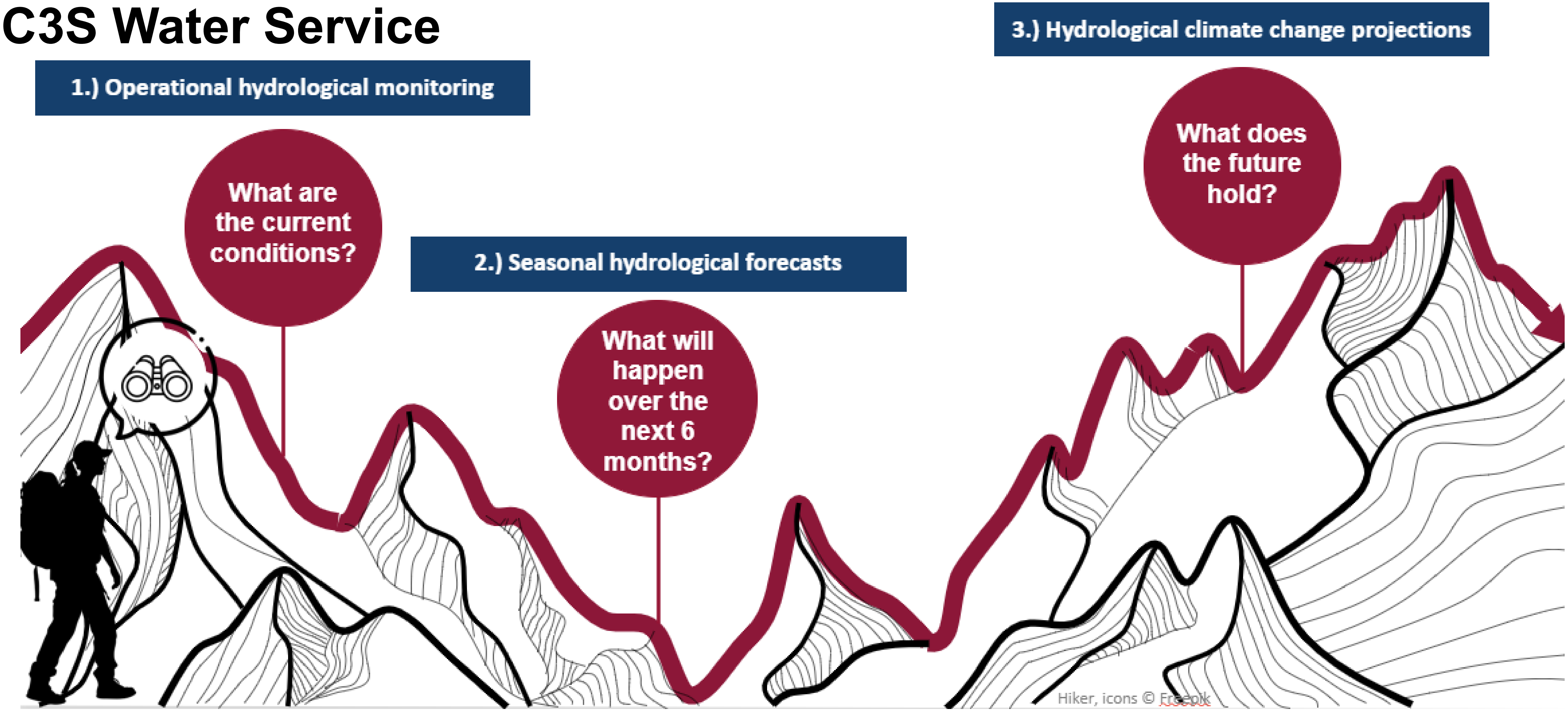
[Operational Seasonal Forecasting]

C3S Water Service



[Operational Seasonal Forecasting]

C3S Water Service

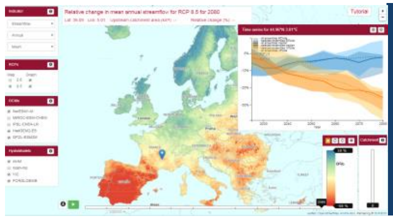
Seasonal forecasts	Projections

- C3S seasonal climate
- CMIP6
- CORDEX
- Consistent regridding & bias adjustment
- Models: mHM, HYPE, ECLand, JULES, VIC, PCR-GlobWB and LISFLOOD
- Outputs: evapotranspiration, snow, soil moisture, river discharge
- Hydrological status
- Six month forecasts
- Hydrological projections
- Climate & water impact indicators

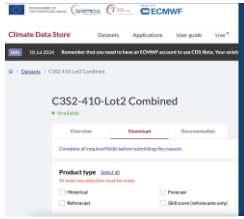
Global *(0.1°)*
Europe *(5km)*

User feedback to shape 2028-2034

Decision support
Strong user engagement
(EIB, EEA, NCP, WMO, …)



Interactive web applications
Data exploration and access for non-technical users



Data on CDS

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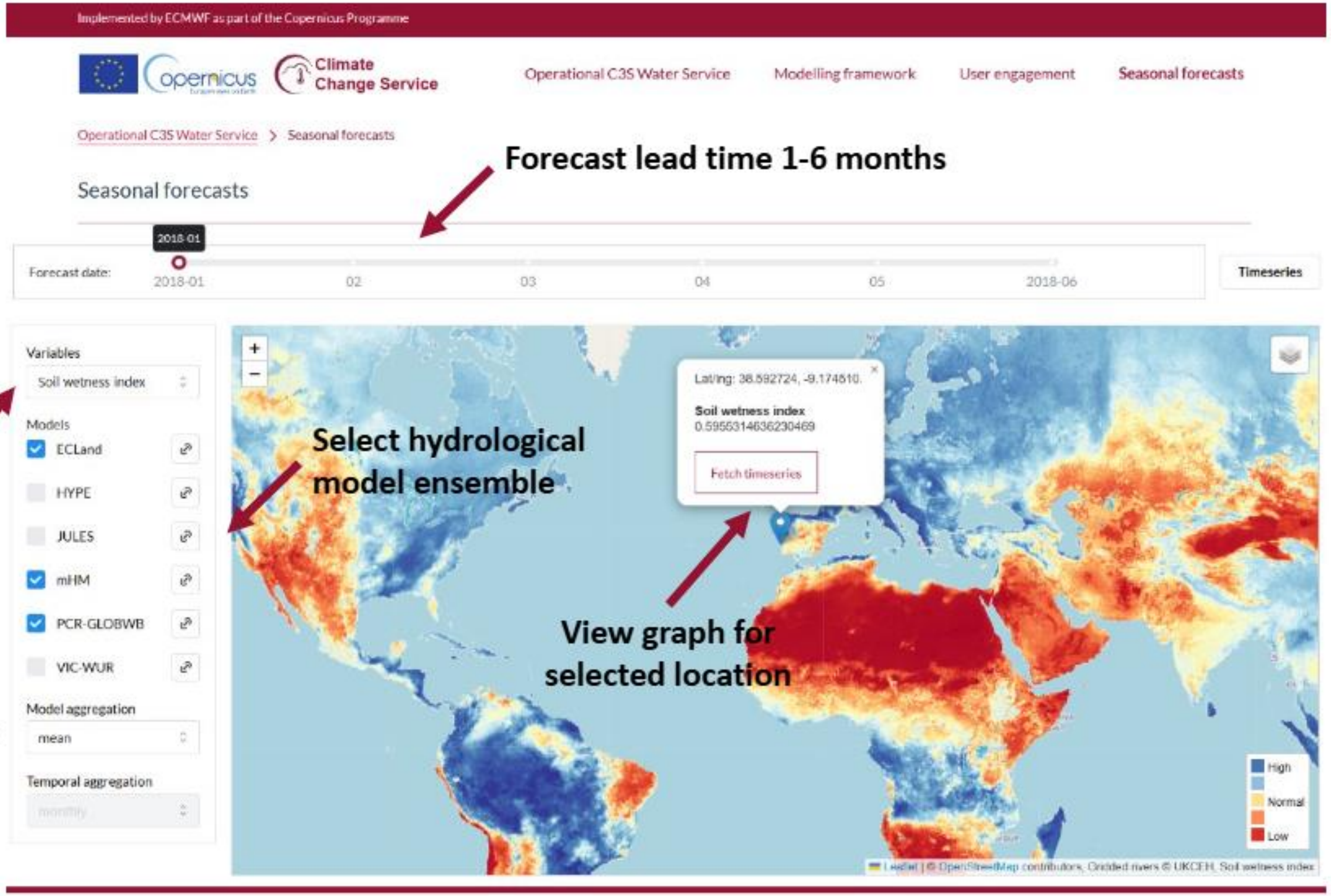
[Operational Seasonal Forecasting]

C3S Water Service

V1 of seasonal forecast app (sample data for 2018)

(not yet publicly available)



Select forecast variable

Select ensemble aggregation / blending approach

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[Operational Seasonal Forecasting]

C3S Water Service – co-developed and responding to user needs

- Stakeholder mapping and identification
- Assess current gaps and needs
- Assess guidance and support documentation requirements
- User-led indicators of future change
- Model ensemble and performance metric needs
- Needs for projections (time slices, warming levels, scenarios…)

[Operational Seasonal Forecasting]

C3S Water Service – co-developed and responding to user needs

Survey to help us better understand your needs for hydrological information from the C3S Water Service

🕒 **~15 minutes**

📋 **Take the survey: https://lnkd.in/df4p7Avu**

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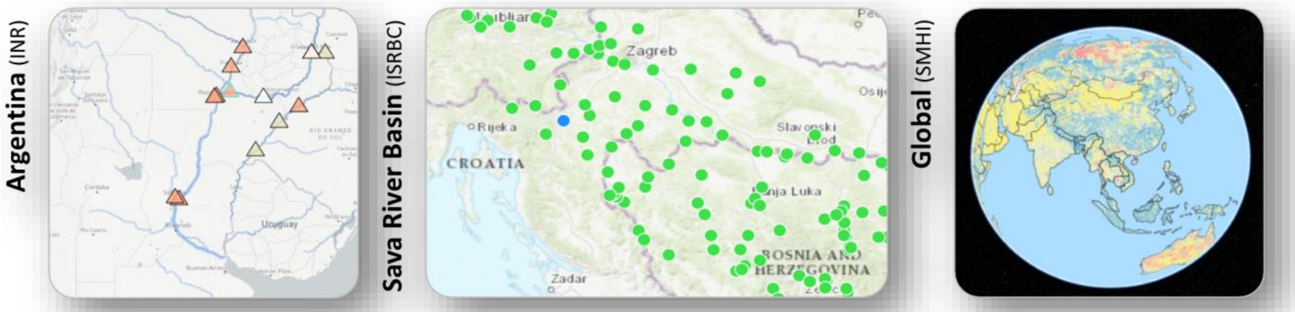
[Operational Seasonal Forecasting]

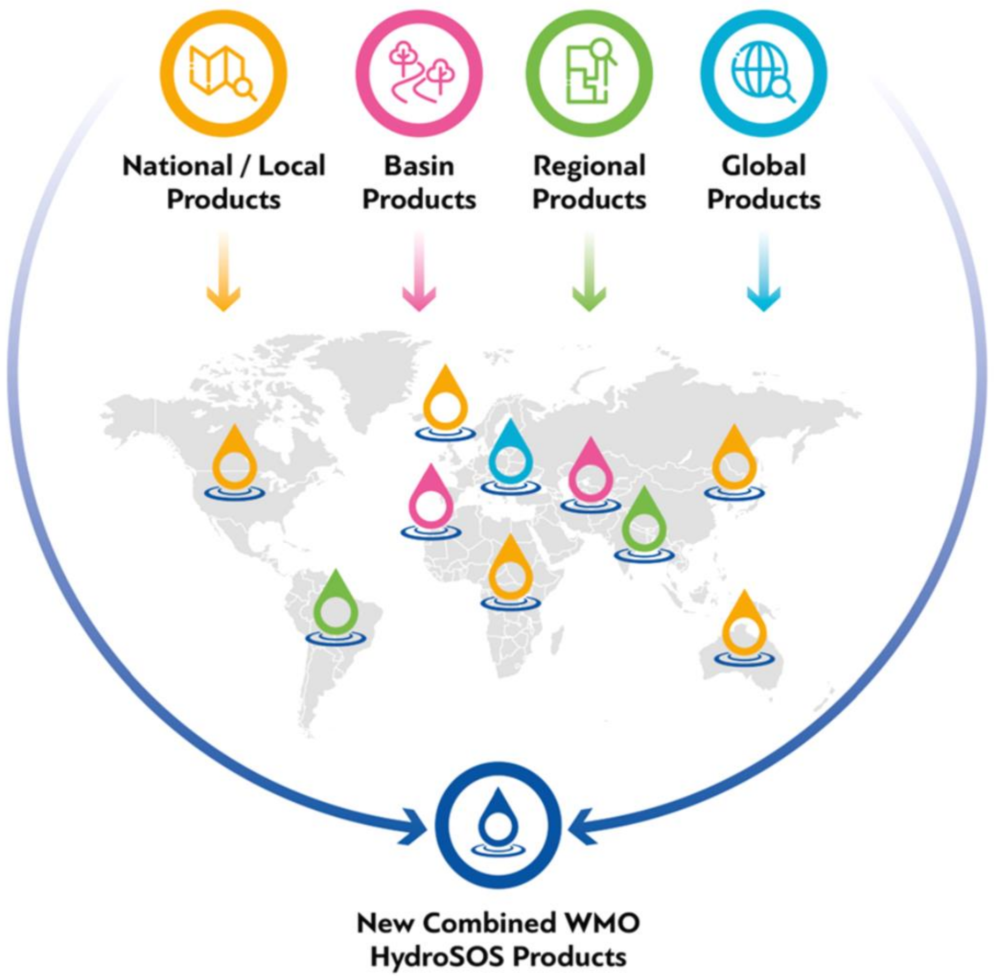
HydroSOS [more this afternoon!]

Connecting products across scales

The disconnect between top-down and bottom-up approaches is an impediment to action



HydroSOS will help enhance services at national, basin, regional and global scales and unite them in a coherent framework



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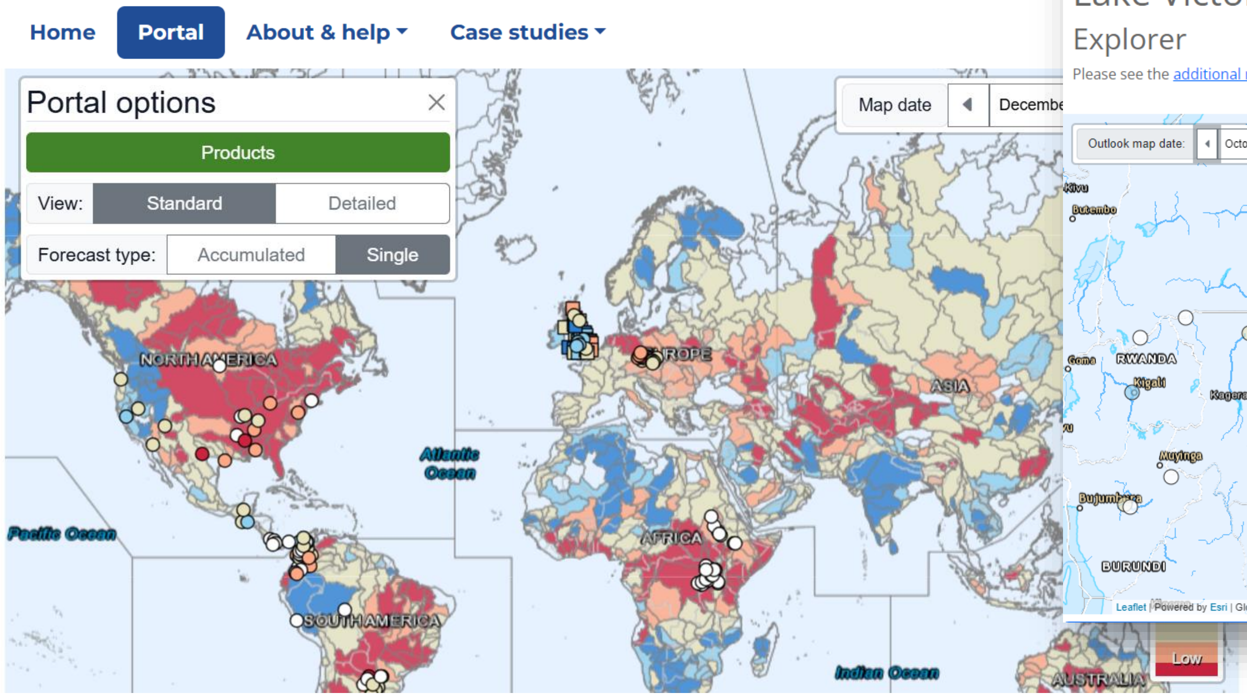
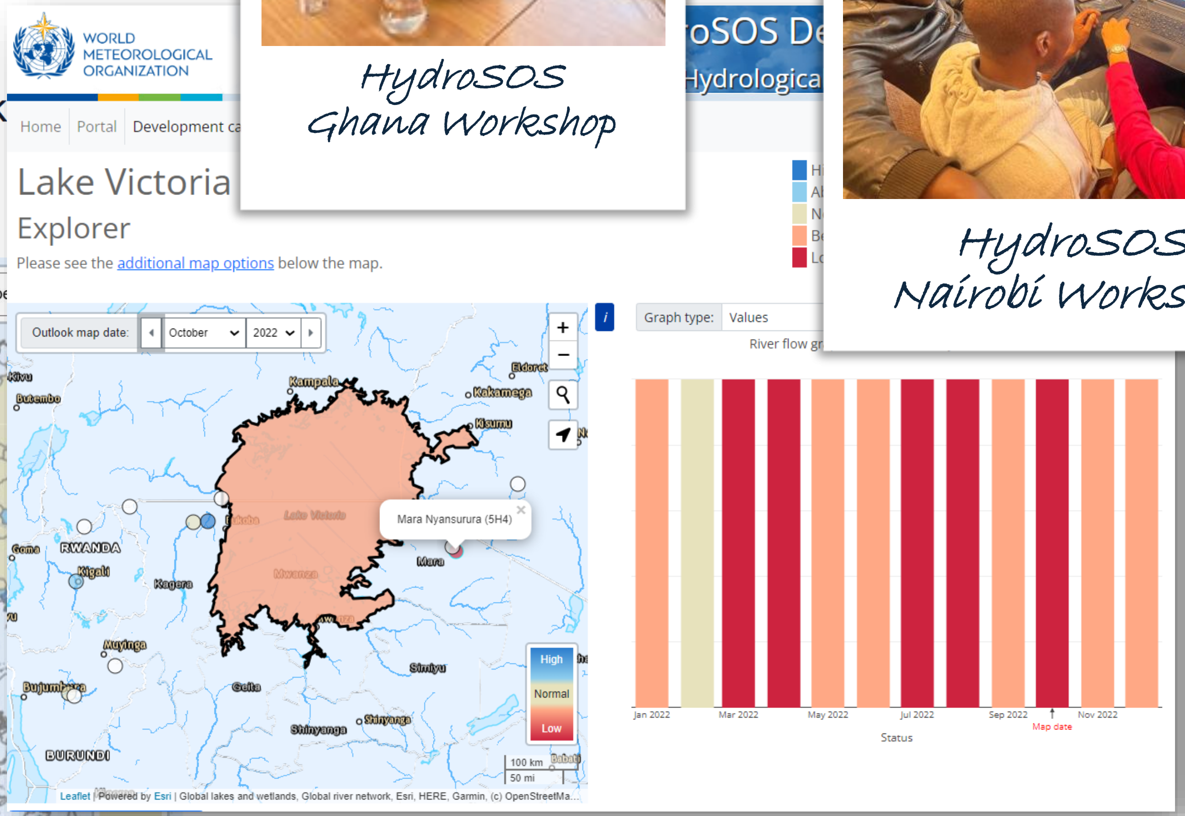
[Operational Seasonal Forecasting]

HydroSOS [more this afternoon!]

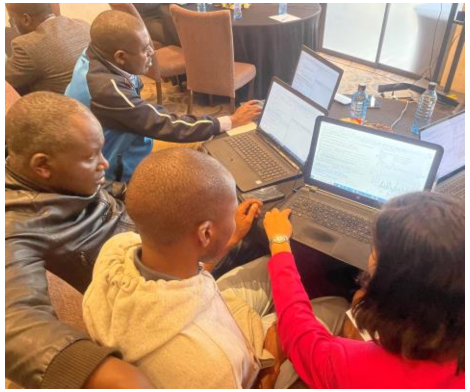
WORLD METEOROLOGICAL ORGANIZATION
Weather Climate Water

WMO Hydrological Status and Outlook
HydroSOS portal





HydroSOS Ghana Workshop



HydroSOS Nairobi Workshop

Thank you

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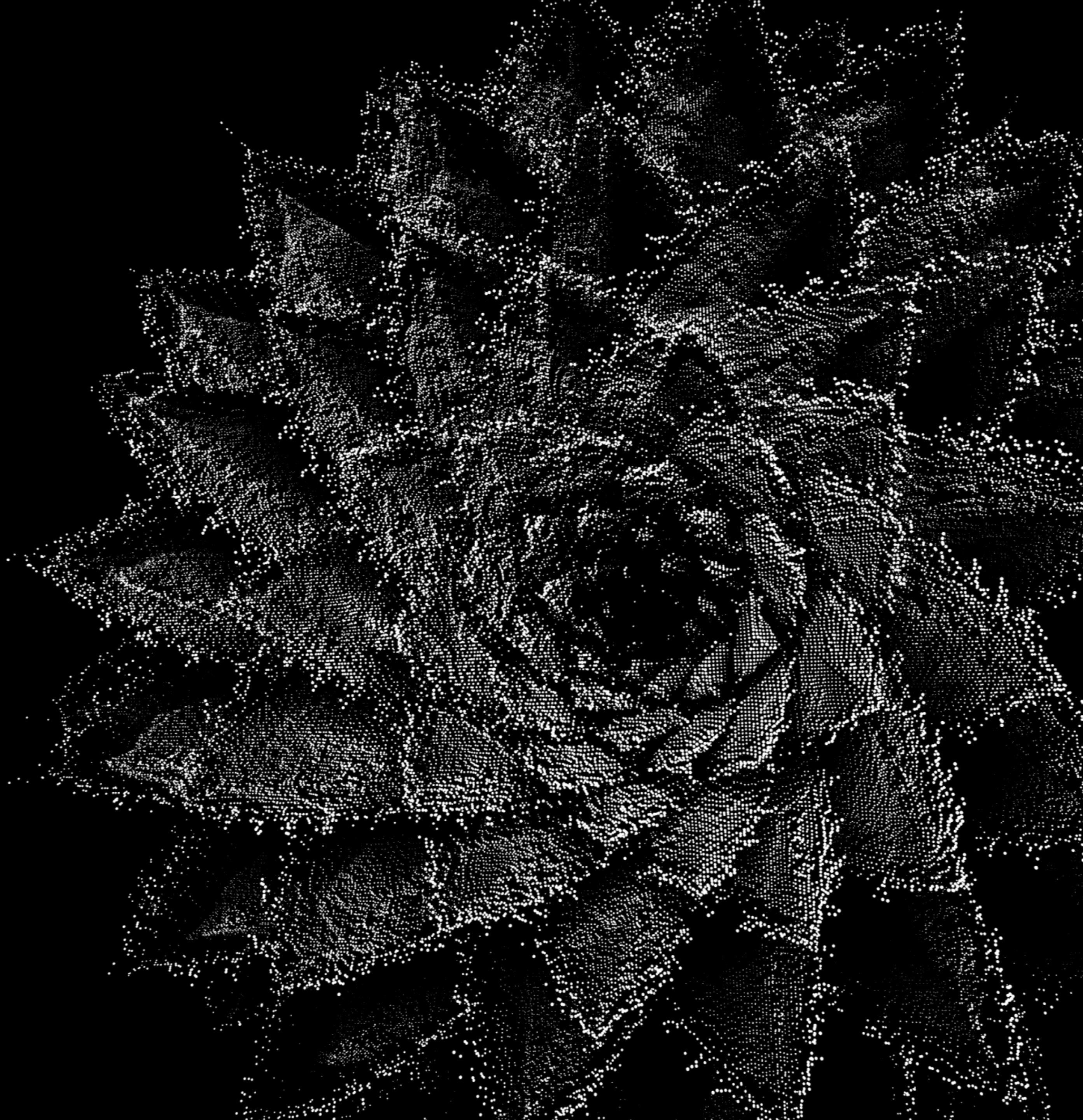


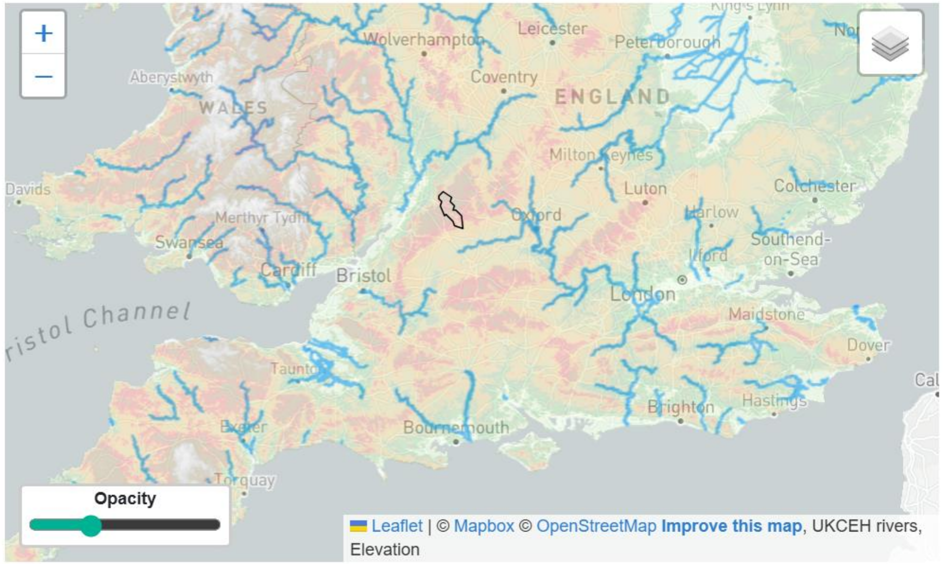
[Interactive Workshop Sessions]

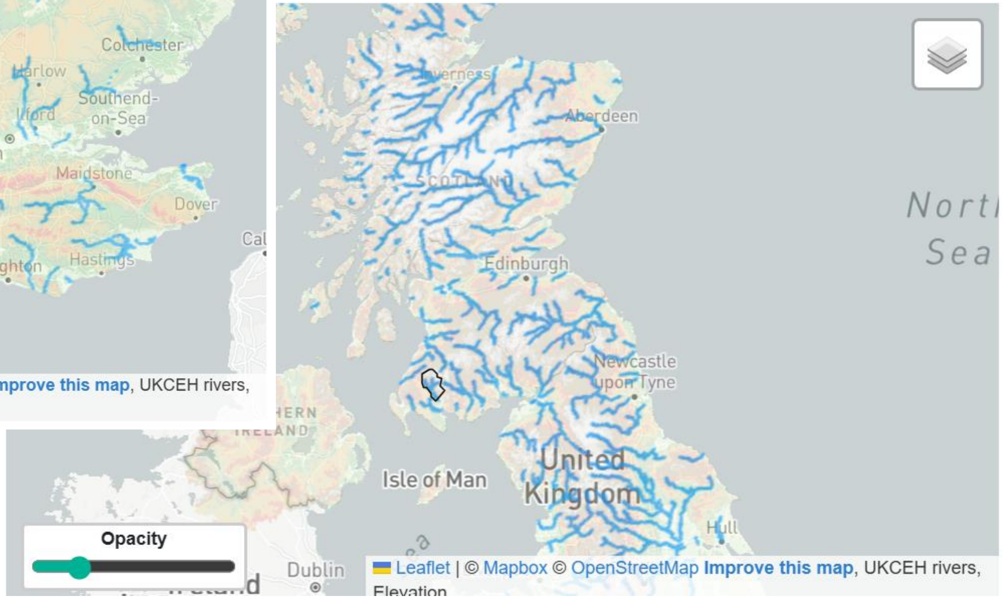
Overview

We'll use the GR6J model to run an Ensemble Streamflow Prediction (ESP) forecast

1) Run GR6J

Calibrate and run for two catchments in the UK: Coln (SE England) and Cree (Scotland)





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[Interactive Workshop Sessions]

Overview

We'll use the GR6J model to run an Ensemble Streamflow Prediction (ESP) forecast

1) Run GR6J
Calibrate and run for two catchments in the UK: Coln (SE England) and Cree (Scotland)

2) Run ESP
Format input data and run ESP forecast for the same two catchments

3) Using forecasts
Format the output data to visualise the forecast